5 Water Quality Impacts

5.1 Introduction

5.1.1 This section presents an assessment of the potential water quality impacts associated with the construction and operation phases of the Hung Shui Kui New Development Area (HSK NDA) (herein referred to as the "Project"). Recommendations for mitigation measures have been provided, where necessary, to minimise the identified water quality impacts to an acceptable level.

5.1.2 An application for an Environmental Permit (EP) would be submitted for the following Schedule 2 Designated Projects (DPs) (refer to **Figure 1.3**) and these DPs have been included in this water quality impact assessment during construction and operational phases:

- Construction of new primary distributor road (Road P1) (DP1)
- Construction of eight new distributor roads (Roads D1 to D8) (DP2)
- Construction of slip roads between: Road D8 Junction and existing Castle Peak Road; Junction of D8/P1 and Junction of D7/P1; and KSWH connection to Road D3 (DP5)
- Construction of partly depressed and partly decked-over roads located at Road D2, Road D4 and Road D6 (DP6)
- Construction of four new SPSs (Sites 2-34, 3-41, 3-48 and 4-35) (DP9)
- Construction of Road P1 and slip-road from KSWH to Road D3 partly located within the "CA" of Yuen Tau Shan (DP12)

5.2 Environmental Legislation, Policies, Plans, Standards and Criteria

Environmental Impact Assessment Ordinance (EIAO)

5.2.1 The Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM) was issued by EPD under Section 16 of the EIAO. It specifies the assessment method and criteria that were followed in this Study. Reference sections in the EIAO-TM provide the details of assessment criteria and guidelines that are relevant to the water quality assessment, including:

- Annex 6 – Criteria for Evaluating Water Pollution; and
- Annex 14 – Guidelines for Assessment of Water Pollution.

Water Quality Objectives

5.2.2 The Water Pollution Control Ordinance (WPCO) provides the major statutory framework for the protection and control of water quality in Hong Kong. According to the Ordinance and its subsidiary legislation, Hong Kong waters are divided into ten Water Control Zones (WCZs). Corresponding statements of Water Quality Objectives (WQO) are stipulated for different water regimes (marine waters, inland waters, bathing beaches subzones, secondary contact recreation subzones and fish culture subzones) in the WCZ based on their beneficial uses. The Project area is located within both Deep Bay and North Western WCZs. The boundaries of Deep Bay and North Western WCZs in relation to the Project area are shown on **Figure 5.1** and **Figure 5.4**. The water quality assessment area for the proposed Project shall cover Deep Bay and North Western WCZs and the corresponding WQOs are listed in **Table 5.1** and **Table 5.2** respectively.

Table 5.1 Summary of Water Quality Objectives for Deep Bay WCZ

Parameters	Criteria	Subzone
Aesthetic appearance	(a) Waste discharges shall cause no objectionable odours or discolouration of the water.	Whole Zone
	(b) Tarry residues, floating wood, articles made of glass, plastic, rubber or of any other substances should be absent.	Whole Zone
	(c) Mineral oil should not be visible on the surface. Surfactants should not give rise to a lasting foam.	Whole Zone
	(d) There should be no recognisable sewage-derived debris.	Whole Zone
	(e) Floating, submerged and semi-submerged objects of a size likely to interfere with the free movement of vessels, or cause damage to vessels, should be absent.	Whole Zone
	(f) Waste discharges shall not cause the water to contain substances which settle to form objectionable deposits.	Whole Zone
Bacteria	(a) The level of *Escherichia coli* should not exceed 610 per 100 mL, calculated as the geometric mean of all samples collected in one calendar year.	Secondary Contact Recreation Subzone and Mariculture Subzone (L.N. 455 of 1991)
	(b) The level of *Escherichia coli* should be zero per 100 ml, calculated as the running median of the most recent 5 consecutive samples taken at intervals of between 7 and 21 days.	Yuen Long & Kam Tin (Upper) Subzone, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones
	(c) The level of *Escherichia coli* should not exceed 1000 per 100 ml, calculated as the running median of the most recent 5 consecutive samples taken at intervals of between 7 and 21 days.	Yuen Long & Kam Tin (Lower) Subzone and other inland waters
	(d) The level of *Escherichia coli* should not exceed 180 per 100 mL, calculated as the geometric mean of all samples collected from March to October inclusive in one calendar year. Samples should be taken at least 3 times in a calendar month at intervals of between 3 and 14 days.	Yuen Long Bathing Beach Subzone (L.N. 455 of 1991)
Colour	(a) Waste discharges shall not cause the colour of water to exceed 30 Hazen units.	Yuen Long & Kam Tin (Upper) Subzone, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones
	(b) Waste discharges shall not cause the colour of water to exceed 50 Hazen units.	Yuen Long & Kam Tin (Lower) Subzone and other inland waters
Dissolved oxygen (DO)	(a) Waste discharges shall not cause the level of dissolved oxygen to fall below 4 milligrams per litre for 90% of the sampling occasions during the year; values should be taken at 1 metre below surface.	Inner Marine Subzone excepting Mariculture Subzone
	(b) Waste discharges shall not cause the level of dissolved oxygen to fall below 4 milligrams per litre for 90% of the sampling occasions during the year; values should be calculated as water column average (arithmetic mean of at least 2 measurements at 1 metre below surface and 1 metre above seabed). In addition, the concentration of dissolved oxygen should not be less than 2 milligrams per litre within 2 metres of the seabed for 90% of the sampling occasions during the year.	Outer Marine Subzone excepting Mariculture Subzone

Parameters	Criteria	Subzone
	(c) The dissolved oxygen level should not be less than 5 milligrams per litre for 90% of the sampling occasions during the year; values should be taken at 1 metre below surface.	Mariculture Subzone
	(d) Waste discharges shall not cause the level of dissolved oxygen to be less than 4 milligrams per litre.	Yuen Long & Kam Tin (Upper and Lower) Subzones, Beas Subzone, Indus Subzone, Ganges Subzone, Water Gathering Ground Subzones and other inland waters of the Zone
pH	(a) The pH of the water should be within the range of 6.5-8.5 units. In addition, waste discharges shall not cause the natural pH range to be extended by more than 0.2 units.	Marine waters excepting Yung Long Bathing Beach Subzone
	(b) Waste discharges shall not cause the pH of the water to exceed the range of 6.5-8.5 units.	Yuen Long & Kam Tin (Upper and Lower) Subzones, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones
	(c) The pH of the water should be within the range of 6.0-9.0 units.	Other inland waters
	(d) The pH of the water should be within the range of 6.0-9.0 units for 95% of samples. In addition, waste discharges shall not cause the natural pH range to be extended by more than 0.5 units.	Yung Long Bathing Beach Subzone
Temperature	Waste discharges shall not cause the natural daily temperature range to change by more than 2.0 degrees Celsius.	Whole Zone
Salinity	Waste discharges shall not cause the natural ambient salinity level to change by more than 10%	Whole Zone
Suspended solids (SS)	(a) Waste discharges shall neither cause the natural ambient level to be raised by 30% nor give rise to accumulation of suspended solids which may adversely affect aquatic communities.	Marine waters
	(b) Waste discharges shall not cause the annual median of suspended solids to exceed 20 milligrams per litre.	Yuen Long & Kam Tin (Upper and Lower) Subzones, Beas Subzone, Ganges Subzone, Indus Subzone, Water Gathering Ground Subzones and other inland waters
Ammonia	The un-ionised ammoniacal nitrogen level should not be more than 0.021 milligram per litre, calculated as the annual average (arithmetic mean).	Whole Zone
Nutrients	(a) Nutrients shall not be present in quantities sufficient to cause excessive or nuisance growth of algae or other aquatic plants.	Inner and Other Marine Subzone
	(b) Without limiting the generality of objective (a) above, the level of inorganic nitrogen should not exceed 0.7 milligram per litre, expressed as annual mean.	Inner Marine Subzone
	(c) Without limiting the generality of objective (a) above, the level of inorganic nitrogen should not exceed 0.5 milligram per litre, expressed as annual water column average (arithmetic mean of at least 2 measurements at 1 metre below surface and 1 metre above seabed).	Outer Marine Subzones

Parameters	Criteria	Subzone
5-day biochemical oxygen demand (BOD)	(a) Waste discharges shall not cause the 5-day biochemical oxygen demand to exceed 3 milligrams per litre.	Yuen Long & Kam Tin (Upper) Subzone, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones
	(b) Waste discharges shall not cause the 5-day biochemical oxygen demand to exceed 5 milligrams per litre.	Yuen Long & Kam Tin (Lower) Subzone and other inland waters
Chemical oxygen demand (COD)	(a) Waste discharges shall not cause the chemical oxygen demand to exceed 15 milligrams per litre.	Yuen Long & Kam Tin (Upper) Subzone, Beas Subzone, Indus Subzone, Ganges Subzone and Water Gathering Ground Subzones
	(b) Waste discharges shall not cause the chemical oxygen demand to exceed 30 milligrams per litre.	Yuen Long & Kam Tin (Lower) Subzone and other inland waters
Toxins	(a) Waste discharges shall not cause the toxins in water to attain such levels as to produce significant toxic carcinogenic, mutagenic or teratogenic effects in humans, fish or any other aquatic organisms, with due regard to biologically cumulative effects in food chains and to toxicant interactions with each other.	Whole Zone
	(b) Waste discharges shall not cause a risk to any beneficial uses of the aquatic environment.	Whole Zone
Phenol	Phenols shall not be present in such quantities as to produce a specific odour, or in concentration greater than 0.05 milligrams per litre as C6H5OH.	Yung Long Bathing Beach Subzone
Turbidity	Waste discharges shall not reduce light transmission substantially from the normal level.	Yung Long Bathing Beach Subzone

Table 5.2 Summary of Water Quality Objectives for North Western WCZ

Parameters	Criteria	Subzone
Aesthetic appearance	(a) Waste discharges shall cause no objectionable odours or discolouration of the water.	Whole Zone
	(b) Tarry residues, floating wood, articles made of glass, plastic, rubber or of any other substances should be absent.	Whole Zone
	(c) Mineral oil should not be visible on the surface. Surfactants should not give rise to a lasting foam.	Whole Zone
	(d) There should be no recognisable sewage-derived debris.	Whole Zone
	(e) Floating, submerged and semi-submerged objects of a size likely to interfere with the free movement of vessels, or cause damage to vessels, should be absent.	Whole Zone
	(f) Waste discharges shall not cause the water to contain substances which settle to form objectionable deposits.	Whole Zone
Bacteria	(a) The level of *Escherichia coli* should not exceed 610 per 100 mL, calculated as the geometric mean of all samples collected in one calendar year.	Secondary Contact Recreation Subzones
	(b) The level of *Escherichia coli* should be less than 1 per 100 mL, calculated as the running median of the most recent 5 consecutive samples taken at intervals of between 7 and 21 days.	Tuen Mun (A) and Tuen Mun (B) Subzones and Water Gathering Ground Subzones

Parameters	Criteria	Subzone
	(c) The level of *Escherichia coli* should not exceed 1000 per 100 mL, calculated as the running median of the most recent 5 consecutive samples taken at intervals of between 7 and 21 days.	Tuen Mun (C) Subzone and other inland waters
	(d) The level of *Escherichia coli* should not exceed 180 per 100 mL, calculated as the geometric mean of all samples collected from March to October inclusive. Samples should be taken at least 3 times in one calendar month at intervals of between 3 and 14 days.	Bathing Beach Subzones
Colour	(a) Waste discharges shall not cause the colour of water to exceed 30 Hazen units.	Tuen Mun (A) and Tuen Mun (B) Subzones and Water Gathering Ground Subzones
	(b) Waste discharges shall not cause the colour of water to exceed 50 Hazen units.	Tuen Mun (C) Subzone and other inland waters
Dissolved oxygen (DO)	(a) Waste discharges shall not cause the level of dissolved oxygen to fall below 4 mg per litre for 90% of the sampling occasions during the whole year; values should be calculated as water column average (arithmetic mean of at least 3 measurements at 1 m below surface, mid-depth and 1 m above seabed). In addition, the concentration of dissolved oxygen should not be less than 2 mg per litre within 2 m of the seabed for 90% of the sampling occasions during the whole year	Marine waters
	(b) Waste discharges shall not cause the level of dissolved oxygen to be less than 4 milligrams per litre.	Tuen Mun (A), Tuen Mun (B) and Tuen Mun (C) Subzones, Water Gathering Ground Subzones and other inland waters
pH	(a) The pH of the water should be within the range of 6.5-8.5 units. In addition, waste discharges shall not cause the natural pH range to be extended by more than 0.2 units.	Marine waters excepting Bathing Beach Subzones
	(b) Waste discharges shall not cause the pH of the water to exceed the range of 6.5-8.5 units.	Tuen Mun (A), Tuen Mun (B) and Tuen Mun (C) Subzones and Water Gathering Ground Subzones
	(c) The pH of the water should be within the range of 6.0-9.0 units.	Other inland waters
	(d) The pH of the water should be within the range of 6.0-9.0 units for 95% of samples. In addition, waste discharges shall not cause the natural pH range to be extended by more than 0.5 units.	Bathing Beach Subzones
Temperature	Waste discharges shall not cause the natural daily temperature range to change by more than 2.0 degrees Celsius.	Whole Zone
Salinity	Waste discharges shall not cause the natural ambient salinity level to change by more than 10%	Whole Zone
Suspended solids (SS)	(a) Waste discharges shall neither cause the natural ambient level to be raised by 30% nor give rise to accumulation of suspended solids which may adversely affect aquatic communities.	Marine waters'
	(b) Waste discharges shall not cause the annual median of suspended solids to exceed 20 milligrams per litre.	Tuen Mun (A), Tuen Mun (B) and Tuen Mun (C) Subzones and Water Gathering Ground Subzones

Parameters	Criteria	Subzone
	(c) Waste discharges shall not cause the annual median of suspended solids to exceed 25 mg per litre.	Other inland waters
Ammonia	The un-ionised ammoniacal nitrogen level should not be more than 0.021 milligram per litre, calculated as the annual average (arithmetic mean).	Whole Zone
Nutrients	(a) Nutrients shall not be present in quantities sufficient to cause excessive or nuisance growth of algae or other aquatic plants.	Marine waters
	(b) Without limiting the generality of objective (a) above, the level of inorganic nitrogen should not exceed 0.3 mg per litre, expressed as annual water column average (arithmetic mean of at least 3 measurements at 1 m below surface, mid-depth and 1 m above seabed).	Castle Peak Bay Subzone
	(c) Without limiting the generality of objective (a) above, the level of inorganic nitrogen should not exceed 0.5 mg per litre, expressed as annual water column average (arithmetic mean of at least 3 measurements at 1 m below surface, mid-depth and 1 m above seabed).	Marine waters excepting Castle Peak Bay Subzone
5-day biochemical oxygen demand (BOD)	(a) Waste discharges shall not cause the 5-day biochemical oxygen demand to exceed 3 milligrams per litre.	Tuen Mun (A), Tuen Mun (B) and Tuen Mun (C) Subzones and Water Gathering Ground Subzones
	(b) Waste discharges shall not cause the 5-day biochemical oxygen demand to exceed 5 milligrams per litre.	Other inland waters
Chemical oxygen demand (COD)	(a) Waste discharges shall not cause the chemical oxygen demand to exceed 15 milligrams per litre.	Tuen Mun (A), Tuen Mun (B) and Tuen Mun (C) Subzones and Water Gathering Ground Subzones
	(b) Waste discharges shall not cause the chemical oxygen demand to exceed 30 milligrams per litre.	Other inland waters
Toxins	(a) Waste discharges shall not cause the toxins in water to attain such levels as to produce significant toxic, carcinogenic, mutagenic or teratogenic effects in humans, fish or any other aquatic organisms, with due regard to biologically cumulative effects in food chains and to toxicant interactions with each other.	Whole Zone
	(b) Waste discharges shall not cause a risk to any beneficial uses of the aquatic environment.	Whole Zone
Phenol	Phenols shall not be present in such quantities as to produce a specific odour, or in concentration greater than 0.05 milligrams per litre as C6H5OH.	Bathing Beach Subzones
Turbidity	Waste discharges shall not reduce light transmission substantially from the normal level.	Bathing Beach Subzones

Technical Memorandum

5.2.3 Discharge of effluents are subject to control under the WPCO. The Technical Memorandum on Standards for Effluents Discharged into Drainage and Sewerage Systems, Inland and Coastal Waters (TM-DSS) gives guidance on the permissible effluent discharges based on the type of receiving waters (foul sewers, stormwater drains, inland and coastal waters). The limits control the physical, chemical and microbial quality of effluents. Any sewage from the proposed construction and operation activities must comply with the standards for effluents discharged into the foul sewers, inland waters and coastal waters of Deep Bay WCZ and North Western WCZ, as given in the TM-DSS.

Practice Note

5.2.4 A Professional Persons Environmental Consultative Committee Practice Note (ProPECC PN) was issued by the EPD to provide guidelines for handling and disposal of construction site discharges. The ProPECC PN 1/94 "Construction Site Drainage" provides good practice guidelines for dealing with 10 types of discharge from construction sites. These include surface run-off, groundwater, boring and drilling water, bentonite slurry, water for testing and sterilisation of water retaining structures and water pipes, wastewater from building constructions, acid cleaning, etching and pickling wastewater, and wastewater from site facilities. Practices given in the ProPECC PN 1/94 should be followed as far as possible during construction to minimise the water quality impact due to construction site drainage.

5.2.5 The ProPECC PN 5/93 "Drainage Plans subject to Comments by Environmental Protection Department" provides guidelines and practices for handling, treatment and disposal of various effluent discharges to stormwater drains and foul sewers. The design of site drainage and disposal of various site effluents generated within the new development area should follow the relevant guidelines and practices as given in the ProPECC PN 5/93.

Technical Circular

5.2.6 Environment, Transport and Works Bureau Technical Circular (ETWB TC) (Works) No. 5/2005 provides an administrative framework to better protect all natural streams/rivers from the impacts of construction works. The procedures promulgated under this Circular aim to clarify and strengthen existing measures for protection of natural streams/rivers from government projects and private developments. The guidelines and precautionary mitigation measures given in the ETWB TC (Works) No. 5/2005 should be followed as far as possible to protect the inland watercourses at or near the Project area during the construction phase.

5.2.7 For maintenance of stormwater drainage system during operational phase, reference should be made to ETWB TC (Works) No. 14/2004 "Maintenance of Stormwater Drainage Systems and Natural Watercourses" where applicable. The circular sets out the departmental responsibilities for the maintenance of stormwater drainage systems and natural watercourses in government and private lands.

5.3 Water Sensitive Receivers for the Project

Existing Inland Watercourses

5.3.1 Major existing watercourses identified within and near the Project area are summarised in **Table 5.3** below and include:

- Tin Shui Wai Main Channel and its tributaries (TSW);
- Hang Hau Tsuen Channel and its tributaries (HTT);
- Small watercourses along Deep Bay (DB);
- Upstream / tributaries of Shan Pui River (SPR); and
- Upstream / tributaries of Tuen Mun River (TMR).

5.3.2 The existing natural and modified watercourses are presented in **Figure 9.1**. In addition to the watercourses presented in **Figure 9.1**, **Figure 5.1** also presents the channelized drainage systems, storm drains, and seasonal or temporary streams observed only under heavy rainfall, in order to provide an overview of existing watercourses(1). The zoom-in maps of these watercourses are provided in **Figures 5.2, 5.2.1** to **5.2.14**.

Tin Shui Wai Main Channel and its Tributaries (TSW)

5.3.3 Tin Shui Wai (TSW) Main Channel and its tributaries are located in the Deep Bay WCZ and are a major fresh water system, which spreads across the majority of the Project area. This channel system generally runs from south to north and eventually enters the marine water at inner Deep Bay. This fresh water system consists of a core channel along the north-eastern Project boundary (namely TSW Main Channel) and three key tributaries (namely Lo Uk Tsuen tributaries, Hung Shiu Kiu Main Street Channel and Tin Sam Channel respectively) as described in **Table 5.3** below and also shown in **Figure 5.1**.

5.3.4 The TSW Main Channel and majority of its key tributary streams would be retained during the operational phase. Revitalisation and greening of some sections including TSW Main Channel, HSK Channel and Tin Sam Channel are proposed (details refer to **Section 5.10.14**). Several minor water sections of this fresh water system will be removed under the Project and their water flow would be diverted to the new covered drainage network within the new development area as summarised in **Table 5.3** below. These minor sections include four small watercourses that drain into Tin Sam Channel (namely TSW1A, TSW1B, TSW1C and TSW1D as shown in **Figure 5.2.9**), 4 minor watercourses near TSW Main Channel (namely TSW2, TSW2A, TSW2B as shown in **Figure 5.2.7** and TSW5 as shown in **Figures 5.2.12 and 5.2.13**) as well as a small section of Lo Uk Tsuen tributaries (namely TSW3 as shown in **Figure 5.2.8**). In addition, a small section (namely TSW4) of Tin Sam Channel will be realigned under this Project as indicated in **Figure 5.2.9**.

Hang Hau Tsuen Channel and its Tributaries (HTT)

5.3.5 Hang Hau Tsuen Channel and its tributaries are comparatively smaller river system found at the northern tip of the Project area as shown in **Figure 5.1**. This fresh water system is also located in the Deep Bay WCZ and it generally runs from south to north and finally enters the marine water at inner Deep Bay.

(1) Source: Base Map Data from Lands Department in 2016 (GeoInfo Map: http://www1.map.gov.hk/gih3/view/index.jsp).

5.3.6 The main stream of Hang Hau Tsuen Channel will be retained during the operational phase. A side branch (namely HHT1) of Hang Hau Channel as shown in **Figure 5.2.1** and the most upstream part (namely HHT2) of Hang Hau Tsuen Channel as shown in **Figure 5.2.7** will be removed under the Project. The water flow of these two affected sections would be diverted to the new covered drainage system within the Project.

Table 5.3 Summary of Major Watercourses at or Near the Project Area

ID (See Figure 5.1)	Existing Fresh Water System	Description				Water Sections to be Removed or Realigned		
		General	Major Sub-sections of the Fresh Water System		Figure Reference (Fig. Ref.)	Name	Description	Fig. Ref.
			Name	Description				
Watercourses Discharging to Deep Bay WCZ								
TSW	TSW Main Channel and its tributaries	A large fresh water system, which spreads across the majority of the Project area and generally runs from south to north and finally enters the marine water of Deep Bay WCZ	TSW Main Channel	Core channel of the fresh water system, which generally runs along the north-eastern boundary of the Project area from south to north and eventually joins the marine water of Deep Bay WCZ in the north.	Overview: Figure 5.1 Close up maps: Figure 5.2.6, Figure 5.2.7 (see upper right) and Figure 5.2.11 (see bottom left)	N/A - The whole section of the Tin Shui Wan Main Channel will be retained under the Project		
			Lo Uk Tsuen tributaries	Upstream tributaries of the fresh water system, which generally run across the middle portion of the Project area from west to east and eventually enters the TSW Main Channel; Some small watercourses (originate from the hill side in the west) also enter these Lo Uk Tsuen tributaries.	Overview: Figure 5.1 Close up maps: Figure 5.2.7 (see bottom right) and Figure 5.2.8 (see upper part)	TSW3	A small upstream section of Lo Uk Tsuen Tributaries to be removed under the Project	Figure 5.2.8
			Hung Shiu Kiu (HSK) Main Street Channel	An upstream tributary of the fresh water system, which originates from a water channel near Wo Ping San Tsuen in the south and then runs towards the north and eventually enters the TSW Main Channel	Overview: Figure 5.1 Close up maps: Figure 5.2.8 (see bottom right) and Figure 5.2.9 (see right hand side)	N/A -The whole section of HSK Main Street Channel will be retained under the Project		
			Tin Sam Channel tributaries	Upstream tributaries of the fresh water system, which originates from a watercourse in the south and then runs near Tin Sam towards the north and eventually joins the HSK Main Street Channel. Some small watercourses (originate from the hillside in the west) also run across the Project area towards the east and then enter Tin Sam Channel	Overview: Figure 5.1 Close up maps: Figure 5.2.4, Figure 5.2.8 (see bottom right) and Figure 5.2.9 (see left hand side)	TSW1A, TSW1B, TSW1C, TSW1D	4 minor sections that finally drain into Tin Sam Channel to be removed under the Project	Figure 5.2.9
						TSW4	A section of Tin Sam Channel to be realigned under the Project	Figure 5.2.9
			Others small tributaries	Small watercourses near Tin Wah Road in the northern Project area, which finally discharge to the TSW Main Channel	Figure 5.2.7 (see upper right)	TSW2, TSW2A, TSW2B	Minor watercourses near Tin Wah Road to be removed under the Project	Figure 5.2.7 (see upper right)

ID (See Figure 5.1)	Existing Fresh Water System	Description				Water Sections to be Removed or Realigned		
		General	Major Sub-sections of the Fresh Water System		Figure Reference (Fig. Ref.)	Name	Description	Fig. Ref.
			Name	Description				
Watercourses Discharging to Deep Bay WCZ								
				Small watercourse near Hung Uk Tsuen in the eastern portion of the Project area, which finally discharges to the TSW Main Channel	Figure 5.2.12, Figure 5.2.13	TSW5	A water section (west of Hung Uk Tsuen) to be removed under the Project	Figures 5.2.12, 5.2.13
HHT	Hang Hau Tsuen Channel and its tributaries	A relatively small stormwater system at the northern tip of the Project area, which generally runs from south to north and finally discharges to the marine water of Deep Bay WCZ.			Overview: Figure 5.1 Close up maps: Figure 5.2.1 and Figure 5.2.7 (see upper middle)	HHT1	A side branch of the Channel to be removed	Figure 5.2.1
						HHT2	An upstream section of the Channel to be removed	Figure 5.2.7
DB	Small water-courses along Deep Bay	These small watercourses are located at the north-western boundary of the Project area, generally running from south to north and finally enters the marine water of Deep Bay WCZ.			Overview: Figure 5.1 Close up map: Figure 5.2.2	N/A – All of these minor watercourses will be retained under the Project		
SBR	Upstream tributaries of Shan Pui River	These small watercourses are located around the proposed combined flushing water service reservoir (FLWSR) and fresh water service reservoir (FWSR) in the south, within the catchment of Shan Pui River, which would eventually discharge to the marine water of Deep Bay WCZ.			Overview: Figure 5.1 Close up map: Figure 5.2.14	N/A – All these minor watercourses will be retained under the Project		
Watercourses discharging to North Western WCZ								
TMR	Upstream tributaries of Tuen Mun River	These watercourses are the upstream tributaries of Tuen Mun River found at the southern tip of the Project area. They are running generally from north to south and eventually enters the marine water of North Western WCZ			Overview: Figure 5.1 Close up map: Figure 5.2.5	TMR1	An upstream section of this tributary to be removed under the Project	Figure 5.2.5

N/A – Not Applicable

Small Watercourses Along Deep Bay (DB)

5.3.7 There are some small watercourses in the northern Project area along the coastline of Deep Bay to the south of Hang Hau Tsuen Channel as shown in **Figure 5.1** and also summarised in **Table 5.3** above. These watercourses generally run from the south to north and finally enter the marine water of Deep Bay WCZ. There will be no change to these minor watercourses during the operational phase.

Upstream Tributaries of Shan Pui River (SMR)

5.3.8 There are also some small watercourses identified around the proposed combined FLWSR and FWSR in the south as shown in **Figure 5.1** and also summarised in **Table 5.3** above. These minor watercourses are actually located within the catchment of Shan Pui River and thus would eventually enter Shan Pui River and discharge into the marine water of Deep Bay WCZ. There will be no change to these minor watercourses during the operational phase.

Upstream Tributaries of Tuen Mun River (TMR)

5.3.9 The watercourses found at the southern tip of the Project area (as shown in **Figure 5.1**) are actually the upstream tributaries of Tuen Mun River. These watercourses are located within the North Western WCZ and they generally run from north to south and eventually to the main channel of Tuen Mun River and discharges to the marine water of the North Western WCZ as indicated in **Table 5.3** above.

5.3.10 The most upstream part (namely TMR1) of the Tuen Mun tributaries, located within the new development area as shown in **Figure 5.2.5**, will be removed under the Project. The original catchment area of this small watercourse (namely TMR1) would form part of the Project. The stormwater originally collected by TMR1 would be collected by the new covered drainage system of the Project. All the remaining downstream sections of the Tuen Mun River will be retained during the operational phase.

Planned Inland Watercourses

5.3.11 No new inland watercourse is proposed under the Project.

Existing Ponds and Wet Agricultural Areas

5.3.12 Other inland water bodies identified at or near the Project area are relatively minor, which include small ponds and wet agricultural areas. A summary of all ponds and wet agricultural areas identified within the assessment area is provided in **Table 5.4** below. A total of 13 ponds (namely P8A, P8B, P8C, P9A, P9B, P9C, P9D, P9E, P9F, P9G, P10, P10C and P12 respectively, which included flood protection, inactive fishponds, wetland mitigation ponds) and five wet agricultural areas (namely A2A, A2B, A2C, A2D and A3A respectively) are located within the Project area. Four ponds found within the existing "Village Type Development" ("V") areas (namely P8A, P9A, P9B and P10C respectively), which form part of the polder schemes (2) for flood protection of the villages, and one reinforced concrete flood storage pond (namely P12) found to the immediate west of KSWH are connected to nearby drainage system / watercourses. All the remaining ponds and wet agricultural areas are isolated from the nearby watercourses and could dry up occasionally after periods of no rain. No particular geographical clustering of these ponds was observed.

(2) The polder schemes include perimeter bunds to prevent the ingress of flood water into the low-lying villages and pumping stations to discharge the flood water within the bund to the nearby drainage system. P8A, P9A, P9B and P10C are part of the polder schemes to retain run-off from flowing into the low-lying village areas.

Table 5.4 Ponds and Agricultural Wet Areas within Assessment Area

ID	Description	Location	Within Project Area? (Yes/No)	Figure Reference	Remarks
P1A	Inactive fish pond	North of Project Area to the south of Hang Hau Tsuen Channel	No	Figure 5.2.1	
P1B	Inactive fish pond	North of Project Area to the south of Hang Hau Tsuen Channel	No	Figure 5.2.1	
P1C	Inactive fish pond	North of Project Area to the south of Hang Hau Tsuen Channel	No	Figure 5.2.1	
P3A	Active fish pond	North of Project area along the coastline of Deep Bay	No	Figure 5.2.2	
P3B	Active fish pond	North of Project area along the coastline of Deep Bay	No	Figure 5.2.2	
P3C	Inactive fish pond	North of Project area along the coastline of Deep Bay	No	Figure 5.2.2	
P3D	Active fish pond	North of Project area along the coastline of Deep Bay	No	Figure 5.2.2	
P6A	Inactive fish pond	South of Project area near the upstream tributaries of Tuen Mun River	No	Figure 5.2.5	
P6B	Inactive fish pond	South of Project area near the upstream tributaries of Tuen Mun River	No	Figure 5.2.5	
P6C	Inactive fish pond	South of Project area near the upstream tributaries of Tuen Mun River	No	Figure 5.2.5	
P6D	Inactive fish pond	South of Project area near the upstream tributaries of Tuen Mun River	No	Figure 5.2.5	
P7	Inactive fish pond	North of Project Area to the west of TSW Main Channel	No	Figure 5.2.6	
P7A	Recreational pond	North of Project Area to the west of TSW Main Channel	No	Figure 5.2.6	
P8A	Flood protection pond	At the "Village Type Development" ("V") near Sik Kong Tsuen	Yes	Figure 5.2.7	
P8B	Inactive fish pond	At the northern part of Project area near San Wai	Yes	Figure 5.2.7	P8B will be removed under the Project
P8C	Inactive fish pond	At the northern part of Project area near San Wai	Yes	Figure 5.2.7	P8C will be removed under the Project
P9A	Flood protection pond	At the "V" in middle portion of the Project area	Yes	Figure 5.2.8	
P9B	Flood protection pond	At the "V" in middle portion of the Project area	Yes	Figure 5.2.8	
P9C	Flood protection pond	At the "V" in middle portion of the Project area	Yes	Figure 5.2.8	
P9D	Wetland mitigation pond	Near the western boundary of Project area underneath KSWH	Yes	Figure 5.2.8	

ID	Description	Location	Within Project Area? (Yes/No)	Figure Reference	Remarks
P9E	Wetland mitigation pond	Near the western boundary of Project area underneath KSWH	Yes	Figure 5.2.8	
P9F	Wetland mitigation pond	West of Project area underneath KSWH	Yes	Figure 5.2.8	
P9G	Wetland mitigation pond	West of Project area underneath KSWH	Yes	Figure 5.2.8	
P9H	Inactive fish pond	West of Project area and to the west of KSWH	No	Figure 5.2.8	
P9I	Inactive fish pond	West of Project area and to the west of KSWH	No	Figure 5.2.8	
P9J	Inactive fish pond	West of Project area and to the west of KSWH	No	Figure 5.2.8	
P10	Inactive fish pond	At the southern portion of Project area near San Sang San Tsuen	Yes	Figure 5.2.9	P10 will be removed under the Project
P10 A	Recreational pond	At "V" east of Project area	No	Figure 5.2.12	
P10 B	Recreational pond	At "V" east of Project area	No	Figure 5.2.12	
P10 C	Flood protection pond	At the "V" in the eastern portion of the Project area	Yes	Figure 5.2.12	
P11 A	Inactive fish pond	North of Yuen Long Highway and to the south of Project area	No	Figure 5.2.13	
P11 B	Inactive fish pond	North of Yuen Long Highway and to the south of Project area	No	Figure 5.2.13	
P12	Reinforced concrete flood storage pond	To the immediate west of KSWH	Yes	Figure 5.2.8	
A1	Wet agricultural area	West of Project area and to the west of KSWH	No	Figure 5.2.8	
A2A	Wet agricultural area	At the "V" in middle portion of the Project area	Yes	Figure 5.2.8	
A2B	Wet agricultural area	At the border of "V" area in middle portion of the Project area	Yes	Figure 5.2.8	Part of A2B, A3C, A3D (located outside the "V" area) will be removed
A2C	Wet agricultural area	At the border of "V" area in middle portion of the Project area	Yes	Figure 5.2.8	
A2D	Wet agricultural area	At the border of "V" area in middle portion of the Project area	Yes	Figure 5.2.8	
A3	Wet agricultural area	South of Project area near the upstream tributaries of Tuen Mun River	No	Figure 5.2.5	
A3A	Wet agricultural area	At the southern part of the Project area near the upstream tributaries of Tuen Mun River	Yes	Figure 5.2.5	A3A will be removed under the Project
A3C	Wet agricultural area	South of Project area near the upstream tributaries of Tuen Mun River	No	Figure 5.2.5	
A4A	Wet agricultural area	North of Yuen Long Highway and to the south of Project area	No	Figure 5.2.13	

ID	Description	Location	Within Project Area? (Yes/No)	Figure Reference	Remarks
A4B	Wet agricultural area	North of Yuen Long Highway and to the south of Project area	No	Figure 5.2.13	
A4C	Wet agricultural area	North of Yuen Long Highway and to the south of Project area	No	Figure 5.2.13	

Shaded cell: Ponds / wet agricultural areas to be completed or partially removed under the Project

5.3.13 Due to the proposed developments, three existing ponds (namely P8B, P8C and P10 respectively as shown in **Figures 5.2.7** and **5.2.9**) and one wet agricultural area (namely A3A as shown in **Figures 5.2.5**) will be completely removed under this Project. In addition, three wet agricultural areas (namely A2B, A2C and A2D respectively as shown in **Figure 5.2.8**) are located at the border of the existing "V" area in middle portion of the Project area. Part of these three wet agricultural areas that are located outside the existing "V" area will be removed under the Project.

Planned Ponds and Agricultural Wet Areas

5.3.14 No new pond / wet agricultural wet area is proposed under the Project.

Planned Flood Retention Facilities

5.3.15 Flood retention facilities which comprise of flood retention lakes and underground storage tanks are proposed under the Project as flood mitigation measures as shown in **Figure 5.3**. The flood retention lakes would be open lakes which are designed for temporarily retaining stormwater run-off to avoid overflow of adjacent drainage channels. The underground storage tanks are covered tanks would be designed for collecting run-off from the nearby drainage system through gravity flow to avoid overflow of nearby open drainage channels. All flood retention facilities will pump the stored run-off to the adjacent drainage channels after heavy rains.

5.3.16 The locations of the flood retention facilities presented in this EIA are indicative and for illustration purpose only. The number and sizes of the flood retention facilities will be subject to further technical review.

Marine Water Sensitive Receivers

5.3.17 The key marine Water Sensitive Receivers (WSRs) within the North Western and Deep Bay WCZs have been identified as follows:

- Existing and planned cooling water intakes along the coastlines of outer Deep Bay, Tuen Mun District and North Lantau;
- Existing and planned flushing water intakes along the coastlines of Tuen Mun District and North Lantau;
- Existing bathing beaches in Tuen Mun District;
- Existing Tuen Mun typhoon shelter;
- Existing Gold Coast marina;
- Planned marina at Tung Chung East;
- Existing Tai O estuary;
- Existing Sha Chau and Lung Kwu Chau Marine Park;

- Planned The Brothers Marine Park;
- Existing Mai Po Inner Deep Bay Ramsar Site;
- Existing oyster culture area at inner Deep Bay;
- Existing fish spawning ground north of Lantau;
- Existing artificial reefs near airport and Sha Chau;
- Other existing ecological resources along the coastlines of Deep Bay, Sha Chau, Lung Kwu Chau, The Brothers and north Lantau including coral habitats, horseshoe crab habitats, seagrass habitats, mangrove habitats and Sites of Special Scientific Interest (SSSIs).

5.3.18 The indicative locations of these major marine WSRs are shown in **Figure 5.4**.

5.4 Water Sensitive Receivers for Sewage Pumping Stations (DP9)

5.4.1 Locations of the four proposed SPS and the tentative alignments of their emergency overflow bypass culverts in relation to the WSRs are shown in **Figure 5.5**. **Table 5.5** below summarises the tentative discharge point of emergency overflow for each SPS and the downstream receiving water that would potentially be affected by the emergency overflow. Detailed description of the receiving water is provided in **Table 5.3** above. Further discussion on the emergency overflow and the receiving water are also given in **Sections 5.11.16** to **5.11.25** below.

Table 5.5 Tentative Discharge Points of Emergency Overflow from SPS and Downstream Receiving Water

SPS (see Figure 5.5)	Discharge Point of Emergency Overflow and Downstream Receiving Water (refer to Table 5.3 for detailed description of Receiving Water)
SPS No. 1	Tin Sam Channel, which would then discharge to the TSW Main Channel and finally to the marine water of Deep Bay WCZ.
SPS No. 2	A side branch of Tin Sam Channel, which would firstly drain into Tin Sam Channel and then discharge to the TSW Main Channel and finally to the marine water of Deep Bay WCZ.
SPS No. 3	Tin Sam Channel, which would then discharge to the TSW Main Channel and finally to the marine water of Deep Bay WCZ.
SPS No. 4	Lo Uk Tsuen Tributaries, which would then enter the TSW Main Channel and finally to the marine water of Deep bay WCZ.

5.5 Baseline Conditions – River Water Quality in Deep Bay WCZ

Tin Shui Wai Main Channel and Its Tributaries (TSW)

5.5.1 The water quality of TSW Main Channel and its tributaries is routinely monitored by EPD. According to the EPD's publication "River Water Quality in Hong Kong in 2014" (which contains the latest information published by EPD on river water quality at the time of writing), the upstream station of TSW Main Channel (TSR2) located near Wo Ping San Tsuen reached a 95% WQO compliance in 2014, compared with 98% in 2013. The downstream station (TSR1) was 82% in compliance in 2014 as compared with 2013's 83%. The two EPD stations at Tin Sui Wai Catchment (TSR1 and TSR2 as shown in **Figure 5.1**) received 'Fair' and 'Good' grading respectively in 2014, similar to 2013. The water quality at the two EPD monitoring stations in the TSW Catchment is summarised in **Table 5.6**.

Table 5.6 Summary Statistics of River Water Quality Data at Tin Shui Wai Catchment Collected by EPD in 2014

Parameters	EPD Stations (Figure 5.1)		WPCO WQO
	TSR1	TSR2	
Dissolved oxygen (DO) (mg/L)	5.8	9.4	Waste discharges shall not cause the level of dissolved oxygen to be less than 4 mg/L
	(3.1 – 9.3)	(8.2 – 12.0)	
pH	7.6	8.2	The pH of the water should be within the range of 6.0-9.0
	(7.2 – 8.0)	(7.4 – 8.7)	
Suspended solids (mg/L)	5	5	Waste discharges shall not cause the annual median of suspended solids to exceed 20mg/L
	(2 - 23)	(2 -8)	
5-day Biochemical Oxygen Demand (BOD) (mg/L)	6	2	Waste discharges shall not cause the 5-day biochemical oxygen demand to exceed 5mg/L
	(4 - 40)	(<1 - 7)	
Chemical Oxygen Demand (COD) (mg/L)	15	6	Waste discharges shall not cause the chemical oxygen demand to exceed 30mg/L
	(6 - 48)	(<2 - 12)	
Oil & grease (mg/L)	<0.5	<0.5	Not available
	(<0.5 – 2.8)	(<0.5 - <0.5)	
Faecal coliforms (cfu/100mL)	650,000	47,000	Not available
	(75,000 - 7,000,000)	(5,300 - 200,000)	
E. coli (cfu/100mL)	210,000	20,000	Not exceed 1000 per 100 ml, calculated as the running median of the most recent 5 consecutive samples taken at intervals of between 7 and 21 days
	(15,000 – 2,500,000)	(2,800 - 88,000)	
Ammonia-nitrogen (mg/L)	2.20	0.34	Not available
	(0.40 – 4.30)	(0.10 – 1.50)	
Nitrate-nitrogen (mg/L)	0.51	0.77	Not available
	(<0.01 - 1.00)	(0.54 - 0.96)	
Total Kjeldahl nitrogen (mg/L)	3.85	0.69	Not available
	(0.83 - 7.20)	(0.19 – 2.20)	
Ortho-phosphate (mg/L)	0.17	0.05	Not available
	(0.06 - 0.38)	(<0.01 - 0.19)	
Total phosphorus (mg/L)	0.34	0.08	Not available
	(0.10 - 0.69)	(0.03 – 0.30)	
Total sulphide (mg/L)	<0.02	<0.02	Not available
	(<0.02 - 0.02)	(<0.02 - <0.02)	
Aluminium (µg/L)	121	133	Not available
	(67 - 393)	(87 - 191)	
Cadmium (µg/L)	<0.1	<0.1	Not available
	(<0.1 - 0.1)	(<0.1 - <0.1)	
Chromium (µg/L)	1	<1	Not available
	(<1 - 2)	(<1 - 1)	
Copper (µg/L)	3	1	Not available
	(2 - 10)	(<1 - 9)	

Parameters	EPD Stations (Figure 5.1)		WPCO WQO
	TSR1	TSR2	
Lead (µg/L)	1	1	Not available
	(<1 - 6)	(<1 - 5)	
Zinc (µg/L)	26	17	Not available
	(18 - 97)	(<10 - 65)	
Flow (L/s)	Not Measured	46	Not available
		(13 – 210)	

Note:
(1) Data source: River Water Quality in Hong Kong in 2014 (EPD).
(2) Data presented are annual arithmetic means of depth-averaged results except for *E. coli* and faecal coliforms that are annual geometric means.
(3) Data in brackets indicate the ranges.

5.5.2 Additional water quality sampling was also conducted under this EIA to supplement the river water quality information for TSW Catchment. The sampling works were carried out in 2014 at three stations in TSW Catchment, including one downstream station (namely TSR3) at the TSW Main Channel and two upstream stations (namely HT and HSK) at the tributaries of TSW Main Channel near Lo Uk Tsuen and Tin Sam, respectively as shown in **Figure 5.1**. The additional sampling works were conducted in six separate days on 2 April, 30 April, 2 May, 8 December, 10 December and 12 December 2014, respectively. Amongst all the WQO parameters with numerical objectives (i.e. DO, pH, SS, BOD, COD and *E. coli*), only about 50% of the measurement data collected from the sampling programme met the WQOs for all the three sampling stations (TSR3, HT and HSK, respectively). The summary statistics of the water quality data collected at these three stations are shown in **Table 5.7**.

Table 5.7 Summary Statistics of River Water Quality at Tin Shui Wai Catchment Collected under this EIA in 2014

Parameters	Sampling Stations (Figure 5.1)			WPCO WQO (in inland waters)
	TSR3	HT	HSK	
Dissolved oxygen (DO) (mg/L)	5.7	5.6	5.7	Waste discharges shall not cause the level of dissolved oxygen to be less than 4 mg/L
	(4.2 – 7.2)	(2.0 - 8.7)	(2.1 - 10.1)	
pH	8	8.3	8.6	The pH of the water should be within the range of 6.0-9.0
	(7.9 – 8.3)	(8.1 - 8.4)	(7.8 – 9.8)	
Suspended solids (SS) (mg/L)	45	16	11	Waste discharges shall not cause the annual median of suspended solids to exceed 20mg/L
	(18 - 74)	(7 – 30)	(8 – 14)	
5-day Biochemical Oxygen Demand (BOD) (mg/L)	4.9	15.0	14.1	Waste discharges shall not cause the 5-day biochemical oxygen demand to exceed 5mg/L
	(3.6 – 6.1)	(2.3 – 36)	(3.5 – 27)	
Chemical Oxygen Demand (COD) (mg/L)	26	38	32	Waste discharges shall not cause the chemical oxygen demand to exceed 30mg/L
	(4 - 39)	(12 – 68)	(11 - 54)	
Faecal coliforms (cfu/100mL)	7,000	34,000	95,000	Not available
	(1,800 - 50,000)	(1,200 – 420,000)	(7,200 – 410,000)	
E. coli (cfu/100mL)	4,300	20,500	72,000	Not exceed 1000 per 100 ml, calculated as the running median of the most recent 5 consecutive samples taken at intervals of between 7 and 21 days
	(800 – 50,000)	(1,200 – 280,000)	(6,900 – 350,000)	

Parameters	Sampling Stations (Figure 5.1)			WPCO WQO (in inland waters)
	TSR3	HT	HSK	
Ammonia-nitrogen (mg/L)	4	3	7	Not available
	(3 – 6)	(<0.01 – 6)	(1 – 13)	
Nitrate-nitrogen (mg/L)	0.32	0.72	0.75	Not available
	(0.14 – 0.61)	(0.09 – 1.5)	(0.24 – 1.30)	
Total Kjeldahl nitrogen (mg/L)	4.7	4.5	7.6	Not available
	(4 - 6)	(<0.1 – 9.9)	(1 – 14)	
Ortho-phosphate (mg/L)	0.26	0.27	0.47	Not available
	(0.14 - 0.41)	(0.07 – 0.51)	(0.07 – 1.1)	
Total phosphorus (mg/L)	0.42	0.5	0.72	Not available
	(0.3 – 0.65)	(0.09 – 0.97)	(0.13 – 1.4)	
Total sulphide (mg/L)	<0.1	<0.1	<0.1	Not available
	(<0.1 - <0.1)	(<0.1 - <0.1)	(<0.1 - <0.1)	
Aluminium (µg/L)	627	74	63	Not available
	(140 – 1,300)	(31 – 190)	(45 – 81)	
Cadmium (µg/L)	0.14	0.10	<0.1	Not available
	(<0.1 – 0.25)	(<0.1 – 0.12)	(<0.1 - <0.1)	
Chromium (µg/L)	1.8	<1	1.2	Not available
	(<1 – 3.7)	(<1 - <1)	(<1 – 1.9)	
Copper (µg/L)	2.9	2.0	2.9	Not available
	(<1 – 7.2)	(<1 – 3.6)	(<1 – 9.5)	
Lead (µg/L)	2.1	1.1	1.2	Not available
	(<1 - 5)	(<1 – 1.4)	(<1 – 2.3)	
Zinc (µg/L)	31	17	15	Not available
	(21 - 45)	(<10 – 27)	(<10 – 19)	
Velocity (L/s)	0.2	0.2	2.7	Not available
	(0.1 – 0.2)	(0.2 – 0.2)	(2.1 – 3.3)	

Note:
(1) Data source: River Water Sampling conducted on six days in 2014 (on 28 & 30 April, 2 May, 8, 10 & 12 Dec) under this Study.
(2) Data presented are annual arithmetic means of depth-averaged results except for *E. coli* and faecal coliforms that are annual geometric means.
(3) Data in brackets indicate the ranges.

5.5.3 The levels of *E. coli* measured in the TSW Catchment at all the five stations (namely TSR1, TSR2, TSR3, HT and HSK) were moderate to high (with annual geometric mean of 82000, 7500, 4300, 20500 and 72000 cfu/100mL, respectively). The *E. coli* levels discharges from the river were still high, largely because of the remaining livestock farms, expedient connections and unsewered villages in the area. The implementation of the Livestock Waste Control Scheme (LWCS) improved the water quality at the TSW Catchment by controlling the effluent quality from livestock farms. In addition, the implementation of the Schemes for the Voluntary Surrender of Poultry and Pig Farm Licence has already brought about a reduction of livestock waste loading. The North District Sewerage Master Plan and Yuen Long and Kam Tin Sewerage Master Plan have included plans to provide public sewers to most of the unsewered villages, and the river water quality should gradually improve as these schemes are implemented.

Hang Hau Tsuen Channel (HHT)

5.5.4 Water quality sampling was also conducted under this EIA to collect river water quality information for Hang Hau Tsuen Channel in the northern part of the Project area. The sampling works were conducted at one station (namely LFS) within the Project boundary as shown in **Figure 5.1**. The additional sampling works were conducted in six separate days on 28 April, 30 April, 2 May, 8 December, 10 December and 12 December 2014, respectively. Amongst all the WQO parameters with numerical objectives (i.e. DO, pH, SS, BOD, COD and *E. coli*), only about 50% of the measurement data collected from the sampling programme met the WQOs at Hang Hau Tsuen Channel. The summary statistics of the water quality data collected at Hang Hau Tsuen Channel are shown in **Table 5.8**.

Table 5.8 Summary Statistics of River Water Quality at Hang Hau Tsuen Channel Collected under this EIA in 2014

Parameters	Additional Stations (Figure 5.1)	WPCO WQO (in inland waters)
	LFS	
Dissolved oxygen (DO) (mg/L)	5	Waste discharges shall not cause the level of dissolved oxygen to be less than 4 mg/L
	3 – 7.8	
pH	7.9	The pH of the water should be within the range of 6.0-9.0
	7.8 – 8.0	
Suspended solids (SS) (mg/L)	26	Waste discharges shall not cause the annual median of suspended solids to exceed 20mg/L
	(9 – 53)	
5-day Biochemical Oxygen Demand (BOD) (mg/L)	11.6	Waste discharges shall not cause the 5-day biochemical oxygen demand to exceed 5mg/L
	(2.8 – 34)	
Chemical Oxygen Demand (COD) (mg/L)	38	Waste discharges shall not cause the chemical oxygen demand to exceed 30mg/L
	(22 – 54)	
Faecal coliforms (cfu/100mL)	24,000	Not available
	(10,000 – 340,000)	
E. coli (cfu/100mL)	13,000	Not exceed 1000 per 100 ml, calculated as the running median of the most recent 5 consecutive samples taken at intervals of between 7 and 21 days
	(3,000 - 100,000)	
Ammonia-nitrogen (mg/L)	4	Not available
	(<0.01 – 5)	
Nitrate-nitrogen (mg/L)	0.53	Not available
	(0.18 – 0.81)	
Total Kjeldahl nitrogen (mg/L)	4.8	Not available
	(<0.1 – 5.5)	
Ortho-phosphate (mg/L)	0.28	Not available
	(0.08 – 0.53)	
Total phosphorus (mg/L)	0.48	Not available
	(0.32 – 0.37)	
Total sulphide (mg/L)	<0.1	Not available
	(<0.1 - <0.1)	
Aluminium (µg/L)	50	Not available
	(40 – 58)	

Parameters	Additional Stations (Figure 5.1)	WPCO WQO (in inland waters)
	LFS	
Cadmium (µg/L)	0.12	Not available
	(<0.1 – 0.21)	
Chromium (µg/L)	<1	Not available
	(<1 - <1)	
Copper (µg/L)	2.1	Not available
	(<1 – 5.4)	
Lead (µg/L)	1.7	Not available
	(<1 – 2.8)	
Zinc (µg/L)	27	Not available
	(<10 – 41)	
Velocity (L/s)	0.2	Not available
	(0.2 – 0.2)	

Note:
(1) Data source: River Water Sampling conducted on six days in 2014 (on 28 & 30 April, 2 May, 8, 10 &12 Dec) under this Study.
(2) Data presented are annual arithmetic means of depth-averaged results except for *E. coli* and faecal coliforms that are annual geometric means.
(3) Data in brackets indicate the ranges.

5.5.5 The catchment areas of Hang Hau Tsuen Channel cover brownfield sites (e.g. container storage areas) in the south and village developments in the north. Similar to the TSW Catchment, the levels of *E. coli* measured in the Hang Hau Tsuen Channel were high (with annual geometric mean of 13,000 cfu/100mL). It is considered that the river quality of Hang Hau Tsuen Channel is still subject to domestic discharges from unsewered villages or expedient connections / contaminated run-off in the area. With implementation of on-going and planned sewerage projects in the Northwestern New Territories, it is expected that the river water quality in Hang Hau Tsuen Channel should gradually improve in the long term.

Small Watercourses along Deep Bay (DB)

5.5.6 Site inspections at these small watercourses were conducted in September 2015 (wet season) and January 2016 (dry season). It was observed during the site inspections that all of these watercourses were dried out or had a very low flow (with water depth of less than 3 cm). Thus, water sampling at these watercourses was not practical. The water in the small watercourses was observed to be turbid similar to that observed in Hang Hau Tsuen Channel. The catchment characteristics of these watercourses are considered similar to that of Hang Hau Tsuen Channel as discussed above. The lower parts of these small watercourses are running through existing village type developments, whilst the upper parts of these watercourses are located at or near the existing brownfield sites. Thus, these watercourses could also receive sewage from villages and contaminated run-off from existing brownfield sites and the water quality of the discharges (if any) from these watercourses could be similar to that of the Hang Hau Tsuen Channel. With implementation of on-going and planned sewerage projects in the Northwestern New Territories, it is expected that pollution loading to these watercourses would be reduced in the future.

Upstream Tributaries of Shan Pui River (SPR)

5.5.7 Site inspections at these small watercourses were conducted in January 2016 (dry season) and June 2016 (wet season). It was observed during the site inspections that all of these watercourses were dry, rending water sampling not possible. However, the catchments of these watercourses are undeveloped areas. These watercourses would mainly collect run-off from the undeveloped areas with no major pollution source. In dry season, these watercourses are expected to be dry, whereas in the wet seasons, the water flows in these watercourses could be rainwater with low pollution levels.

5.6 Baseline Conditions – River Water Quality in North Western WCZ

Tuen Mun River

5.6.1 The water quality of Tuen Mun River is also routinely monitored by EPD. According to the EPD's publication "River Water Quality in Hong Kong in 2014", Tuen Mun River showed significant improvement in the last decade. Its WQO compliance rate rose steadily from 62% in 1997 to 90% in 2014. All but the upstream station (TN1) were graded 'Excellent' or 'Good' in 2014, similar to 2013. This improvement was attributed to pollution control efforts as well as implementation of the mitigation measures recommended under the Tuen Mun Sewerage Master Plan. However, the upstream station of (TN1) was still 'Bad' and impacted by unsewered villages with an *E. coli* annual geometric mean level of 120,000 cfu/100mL in 2014. The *E. coli* levels of the other stations, namely TN2, TN3, TN4, TN5 and TN6, remained relatively high at 29,000, 4,500, 7,600, 6,500 and 2,400 cfu/100mL respectively. A summary of the water quality data collected at the monitoring stations in the Tuen Mun River is shown in **Table 5.9**.

Table 5.9 Summary Statistics of River Water Quality at Tuen Mun River Collected by EPD in 2014

Parameters	Tuen Mun River						WPCO WQO (in inland waters)
	TN1	TN2	TN3	TN4	TN5	TN6	
Dissolved Oxygen (DO) (mg/L)	4.8	8.5	6.1	6.5	5.8	6.1	Waste discharges shall not cause the level of dissolved oxygen to be less than 4 mg/L.
	(3.1 – 9.9)	(6.1 - 11.7)	(4.2 – 8.7)	(5.1 – 9.3)	(4.8 - 7.7)	(4.1 – 7.3)	
pH	7.7	7.6	7.7	7.7	7.6	7.6	The pH of the water should be within the range of 6.5-8.5 (for TN2) and 6.0-9.0 (for remaining stations).
	(7.5 – 9.2)	(7.2 - 8.3)	(7.3 – 8.2)	(7.5 – 8.3)	(7.4 – 8.1)	(7.0 – 8.1)	
Suspended Solids (SS) (mg/L)	10	4	3	3	3	3	Waste discharges shall not cause the annual median of suspended solids to exceed 20 mg/L (for TN2) and 25 mg/L (for remaining stations).
	(5 - 120)	(<1 - 64)	(1 - 29)	(2 - 12)	(2 - 15)	(2 - 9)	
5-day Biochemical Oxygen Demand (BOD) (mg/L)	24	3	2	2	2	2	Waste discharges shall not cause the 5-day biochemical oxygen demand to exceed 3 mg/L (for TN2) and 5 mg/L (for remaining stations).
	(10 - 38)	(<1 - 9)	(<1 - 6)	(<1 - 7)	(<1 - 6)	(<1 - 4)	
Chemical Oxygen Demand (COD) (mg/L)	35	6	7	9	10	6	Waste discharges shall not cause the chemical oxygen demand to exceed 15 mg/L (for TN2) and 30 mg/L (for remaining stations).
	(18 - 45)	(4 - 31)	(4 - 14)	(5 - 17)	(6 - 16)	(2 - 12)	
Oil & grease (mg/L)	0.8	<0.5	<0.5	<0.5	<0.5	<0.5	Not available
	(<0.5 – 1.3)	(<0.5 - 0.6)	(<0.5 – <0.5)	(<0.5 - <0.5)	(<0.5 - <0.5)	(<0.5 - <0.5)	
Faecal coliforms (cfu/100mL)	540,000	65,000	48,000	59,000	51,000	21,000	Not available
	(94,000 – 2,600,000)	(29,000 - 350,000)	(650 - 450,000)	(7,700 - 300,000)	(8,900 - 260,000)	(190 - 500,000)	
E. coli (cfu/100mL)	120,000	29,000	4,500	7,600	6,500	2,400	Not exceed 1000 per 100 ml, calculated as the running median of the most recent 5 consecutive samples taken at intervals of between 7 and 21 days
	(62,000 - 320,000)	(11,000 - 61,000)	(160 - 31,000)	(1,600 - 50,000)	(900 - 49,000)	(10 - 51,000)	
Ammonia-nitrogen (mg/L)	6.55	1.45	0.36	0.33	0.35	0.41	Not available
	(0.35 – 9.30)	(0.29 - 7.00)	(0.25 – 0.69)	(0.15 – 1.20)	(0.27 – 0.78)	(0.26 - 0.76)	

Parameters	Tuen Mun River						WPCO WQO (in inland waters)
	TN1	TN2	TN3	TN4	TN5	TN6	
Nitrate-nitrogen (mg/L)	0.49	2.00	0.42	0.45	0.47	0.35	Not available
	(<0.01 - 1.60)	(1.30 - 3.00)	(0.24 - 0.94)	(0.23 –.0.96)	(0.21 – 0.95)	(0.14 – 1.20)	
Total Kjeldahl nitrogen (mg/L)	8.85	1.85	0.68	0.65	0.66	0.62	Not available
	(2.70 – 11.00)	(0.51 – 8.90)	(0.52 – 1.10)	(0.28 - 1.60)	(0.47 – 1.20)	(0.43 – 1.20)	
Ortho-phosphate (mg/L)	0.58	0.22	0.04	0.03	0.03	0.04	Not available
	(0.16 - 0.91)	(0.04 - 1.00)	(<0.01 - 0.06)	(0.02 - 0.06)	(0.01 - 0.06)	(0.01 - 0.06)	
Total phosphorus (mg/L)	0.93	0.25	0.07	0.06	0.06	0.06	Not available
	(0.30 - 1.40)	(0.05 – 1.20)	(0.05 - 0.10)	(0.04 - 0.10)	(0.05 - 0.10)	(0.05 - 0.14)	
Total sulphide (mg/L)	0.04	<0.02	<0.02	<0.02	<0.02	<0.02	Not available
	(<0.02 - 0.14)	(<0.02 - <0.02)	(<0.02 - <0.02)	(<0.02 - <0.02)	(<0.02 - 0.02)	(<0.02 - <0.02)	
Aluminium (µg/L)	96	174	144	162	146	80	Not available
	(68 – 1,142)	(92 - 524)	(<50 - 316)	(<50 - 279)	(<50 - 302)	(<50 - 175)	
Cadmium (µg/L)	<0.1	<0.1	<0.1	<0.1	<0.1	<0.1	Not available
	(<0.1 - 0.2)	(<0.1 - 0.1)	(<0.1 - 0.2)	(<0.1 - 0.2)	(<0.1 - 0.1)	(<0.1 - 0.2)	
Chromium (µg/L)	<1	1	3	3	3	3	Not available
	(<1 - 4)	(<1 - 4)	(1 – 5)	(2 - 5)	(2 - 5)	(1 - 6)	
Copper (µg/L)	5	2	7	5	5	6	Not available
	(3 - 69)	(<1 - 10)	(4 - 10)	(2 - 10)	(2 - 11)	(3 - 18)	
Lead (µg/L)	1	2	<1	<1	1	<1	Not available
	(<1 - 14)	(<1 - 12)	(<1 - 3)	(<1 - 4)	(<1 - 3)	(<1 - 3)	
Zinc (µg/L)	36	21	23	19	18	22	Not available
	(20 - 133)	(<10 - 93)	(12 - 62)	(14 - 45)	(11 - 42)	(<10 - 60)	
Flow (L/s)	137	14	Not Measured	Not Measured	Not Measured	Not Measured	Not available
	(23 - 270)	(3 - 102)					

Note:
(1) Data source: River Water Quality in Hong Kong in 2014 (EPD).
(2) Data presented are in annual medians of monthly samples, except those for faecal coliforms and *E.coli* which are in annual geometric means.
(3) Figures in brackets are annual ranges.

5.7 Baseline Conditions – Marine Water Quality in Deep Bay WCZ

5.7.1 Marine water quality monitoring data routinely collected by EPD were used to establish the baseline condition. The EPD monitoring data collected in 2014 were summarised in **Table 5.10** for Inner Deep Bay (Stations DM1 - DM3) and Outer Deep Bay (Stations DM4 and DM5). Indicative locations of these monitoring stations are shown in **Figure 5.4**. Descriptions of the baseline water quality conditions provided in the subsequent sections are extracted from the EPD's report "Marine Water Quality in Hong Kong 2014", which contains the latest information published by EPD on marine water quality at the time of writing.

5.7.2 The Project area is located within the Deep Bay WCZ. Deep Bay is shallow with depths varying from 2 m at DM1 to 8 m at DM5 and its sediment laden water body is impacted by pollution from both Hong Kong and Shenzhen including discharges from Shenzhen River as well as some local unsewered villages. The Inner Deep Bay was most affected by the discharges from Shenzhen River as well as Kam Tin River, Yuen Long Creek and TSW Main Channel. The WQO compliance rate of the Deep Bay WCZ in 2014 was 40%, same as from 2008 to 2013 except in 2012 (53%). Similar to the previous years, the Deep Bay WCZ had high nutrient levels in 2014. The Total Inorganic Nitrogen (TIN) level in the Deep Bay inner subzone (1.5 – 3.5 mg/L) and Deep Bay outer subzone (0.9 – 1.2 mg/L) was higher than the respective TIN objective of 0.7 mg/L and 0.5 mg/L. Non-compliance with the Unionized Ammonia (UIA) objective and DO objective was also observed at two of the three stations in Deep Bay inner subzone (DM1 and DM2).

5.8 Baseline Conditions – Marine Water Quality in North Western Water

5.8.1 A summary of the marine water quality data collected at the monitoring stations in the North Western WCZ is given in **Table 5.11**. Indicative locations of these monitoring stations (namely NM1 – NM8) are shown in **Figure 5.4**. According to the EPD's publication "Marine Water Quality in Hong Kong 2014", the North Western WCZ attained an overall WQO compliance rate of only 61% in 2014 because of lower compliance with the DO objective (50%) due to the hot summer weather. Compliance with TIN objectives in this WCZ was 33%. The higher levels of TIN (annual means 0.42 - 0.74 mg/L) might be associated with the higher background level of the discharges from Pearl River, and other local discharges. All stations (namely NM1 - NM8) in the WCZ fully complied with the UIA objective.

Table 5.10 Summary Statistics of Marine Water Quality of Deep Bay WCZ Collected by EPD in 2014

Parameter		Inner Deep Bay			Outer Deep Bay		WPCO WQOs (in marine waters)
		DM1	DM2	DM3	DM4	DM5	
Temperature (°C)		24.2 (14.3 – 32.1)	24.4 (14.5 - 31.9)	24.4 (15.1 - 32.0)	24.2 (15.4 - 30.7)	24.1 (16.3 - 30.7)	Not more than 2°C in daily temperature range
Salinity		15.5 (3.5 - 24.4)	17.5 (5.6 – 25.8)	21.2 (10.7 - 29.4)	22.7 (14.8 - 30.5)	25.7 (15.3 - 33.0)	Not to cause more than 10% change
Dissolved Oxygen (DO) (mg/L)	Depth average	3.7 (1.3 - 6.1)	4.6 (2.6 – 6.8)	5.5 (4.1 – 7.3)	5.5 (3.8 – 7.4)	5.8 (4.1 – 7.3)	Not less than 4 mg/L for 90% of the samples
	Bottom	Not measured	Not measured	Not measured	5.6 (3.6 – 7.4)	5.7 (3.6 - 7.2)	Not less than 2 mg/L for 90% of the samples
Dissolved Oxygen (DO) (% Saturation)	Depth average	48 (18 - 73)	60 (36 - 81)	73 (60 - 88)	75 (55 - 89)	79 (62 - 94)	-
	Bottom	Not measured	Not measured	Not measured	75 (53 - 91)	78 (53 - 97)	-
pH		7.4 (7.1 - 7.7)	7.5 (7.3 – 7.9)	7.7 (7.5 – 7.9)	7.8 (7.5 - 8.0)	7.9 (7.7 - 8.1)	6.5 - 8.5 (± 0.2 from natural range)
Suspended Solids (mg/L)		46.2 (15.0 - 95.0)	23.0 (13.0 - 41.0)	15.5 (4.9 - 53.0)	12.8 (2.9 - 47.5)	6.3 (2.4 - 10.6)	Not more than 30% increase
Ammonia Nitrogen (NH_3-N) (mg/L)		2.080 (0.930 - 4.500)	1.410 (0.330 - 2.800)	0.536 (0.069 - 1.700)	0.282 (0.019 - 0.850)	0.194 (0.024 - 0.657)	-
Unionised Ammonia (UIA) (mg/L)		0.026 (0.006 - 0.077)	0.025 (0.005 - 0.076)	0.014 (0.001 - 0.045)	0.006 (<0.001 - 0.011)	0.005 (<0.001 - 0.011)	Not more than 0.021 mg/L
Nitrite Nitrogen (NO_2-N) (mg/L)		0.367 (0.170 - 0.560)	0.291 (0.100 - 0.570)	0.184 (0.055 - 0.350)	0.155 (0.036 - 0.360)	0.124 (0.016 - 0.283)	-
Nitrate Nitrogen (NO_3-N) (mg/L)		1.030 (0.380 - 2.500)	0.918 (0.460 - 1.800)	0.759 (0.410 - 1.100)	0.730 (0.245 - 1.100)	0.561 (0.104 - 1.030)	-
Total Inorganic Nitrogen (TIN) (mg/L)		3.48 (2.36 – 5.74)	2.61 (1.56 – 3.75)	1.48 (0.84 - 3.07)	1.17 (0.56 - 2.24)	0.88 (0.24 - 1.82)	Not more than 0.7 mg/L (for Inner Deep Bay and not more than 0.5mg/L (for Outer Deep Bay)
Total Nitrogen (TN) (mg/L)		4.17 (2.43 - 6.80)	3.15 (2.02 – 4.85)	1.79 (1.12 – 3.77)	1.39 (0.78 – 2.59)	1.08 (0.40 – 2.11)	-
Orthophosphate Phosphorus (Ortho P) (mg/L)		0.213 (0.110 - 0.320)	0.183 (0.110 - 0.260)	0.093 (0.045 - 0.180)	0.047 (0.032 - 0.070)	0.033 (0.015 - 0.050)	-

Parameter	Inner Deep Bay			Outer Deep Bay		WPCO WQOs (in marine waters)
	DM1	DM2	DM3	DM4	DM5	
Total Phosphorus (TP) (mg/L)	0.31 (0.18 - 0.50)	0.27 (0.17 - 0.41)	0.13 (0.08 - 0.24)	0.07 (0.05 - 0.10)	0.05 (0.03 - 0.07)	-
Chlorophyll-*a* (µg/L)	6.4 (1.7 - 17.0)	7.6 (1.8 - 31.0)	4.7 (0.4 - 18.0)	1.7 (<0.2 - 5.9)	2.6 (0.4 – 15.7)	-
E. coli (cfu/100 mL)	1300 (72 - 140000)	380 (36 - 7800)	37 (<1 - 840)	90 (19 - 600)	170 (23 - 450)	-

Note:
(1) Data source: Marine Water Quality in Hong Kong in 2014 (EPD).
(2) Except as specified, data presented are depth-averaged values calculated by taking the means of three depths: surface, mid-depth and bottom.
(3) Data presented are annual arithmetic means of depth-averaged results except for *E.coli* and faecal coliforms that are annual geometric means.
(4) Data in brackets indicate the ranges.

Table 5.11 Summary Statistics of Marine Water Quality of North Western Water WCZ Collected by EPD in 2014

Parameter		Lantau Island (North)	Pearl Island	Pillar Point	Urmston Road	Chek Lap Kok		WPCO WQO (in marine waters)
		NM1	NM2	NM3	NM5	NM6	NM8	
Temperature (°C)		23.4 (17.2 - 28.9)	23.9 (17.2 - 29.0)	23.9 (17.3 - 28.9)	24.0 (17.3 - 29.0)	24.3 (17.1 - 30.2)	23.8 (17.1 - 29.0)	Not more than 2°C in daily temperature range
Salinity		30.2 (25.2 - 32.3)	28.3 (19.5 - 32.4)	28.0 (19.7 - 32.4)	27.1 (20.4 - 32.2)	25.3 (10.1 - 32.5)	28.5 (18.6 - 32.9)	Not to cause more than 10% change
Dissolved Oxygen (DO) (mg/l)	Depth average	5.6 (3.2 - 8.5)	5.8 (3.8 - 8.3)	5.8 (3.8 - 9.0)	5.7 (3.9 - 9.8)	6.3 (4.4 - 8.3)	6.2 (4.3 - 9.0)	Not less than 4 mg/l for 90% of the samples
	Bottom	5.4 (1.9 - 9.5)	5.7 (3.2 - 9.5)	5.7 (2.9 - 9.9)	5.6 (2.8 - 9.5)	6.3 (4.2 - 9.2)	5.8 (0.8 - 9.4)	Not less than 2 mg/l for 90% of the samples
Dissolved Oxygen (DO) (% Saturation)	Depth average	77 (46 - 108)	80 (55 - 106)	80 (55 - 114)	77 (57 - 125)	86 (63 - 107)	85 (62 - 115)	-
	Bottom	75 (27 - 121)	79 (47 - 120)	79 (42 - 126)	77 (41 - 121)	86 (61 - 116)	80 (12 - 121)	-
pH		7.9 (7.6 - 8.0)	7.9 (7.5 - 8.1)	7.9 (7.5 - 8.1)	7.8 (7.4 - 8.1)	7.9 (7.4 - 8.1)	8.0 (7.6 - 8.1)	6.5 - 8.5 (± 0.2 from natural range)
Suspended Solids (SS) (mg/l)		6.6 (1.1 - 25.7)	4.3 (1.5 - 12.3)	7.6(2.0 - 30.0)	6.8 (2.8 - 20.7)	8.6 (2.6 - 44.0)	9.0 (2.1 - 26.7)	Not more than 30% increase
Ammonia Nitrogen (NH_3-N) (mg/L)		0.093 (0.031 - 0.176)	0.107 (0.014 - 0.260)	0.108 (0.006 - 0.277)	0.122 (0.005 - 0.297)	0.091 (<0.005 - 0.243)	0.041 (0.006 - 0.094)	-

Parameter	Lantau Island (North)	Pearl Island	Pillar Point	Urmston Road	Chek Lap Kok		WPCO WQO (in marine waters)
	NM1	NM2	NM3	NM5	NM6	NM8	
Unionised Ammonia (UIA) (mg/L)	0.003 (0.001 - 0.006)	0.003 (<0.001 - 0.008)	0.003 (<0.001 - 0.008)	0.004 (<0.001 - 0.008)	0.003 (<0.001 - 0.008)	0.002 (<0.001 - 0.004)	Not more than 0.021 mg/l for annual mean
Nitrite Nitrogen (NO_2-N) (mg/L)	0.058 (0.023 - 0.115)	0.081 (0.024 - 0.233)	0.091 (0.022 - 0.263)	0.108 (0.022 - 0.323)	0.108 (0.016 - 0.270)	0.067 (0.011 - 0.163)	-
Nitrate Nitrogen (NO_3-N) (mg/L)	0.268 (0.067 - 0.603)	0.394 (0.051 - 0.967)	0.426 (0.056 – 0.997)	0.480 (0.047 - 1.010)	0.544 (0.044 - 1.370)	0.364 (0.014 - 1.180)	-
Total Inorganic Nitrogen (TIN) (mg/L)	0.42 (0.15 – 0.71)	0.58 (0.12 - 1.13)	0.63 (0.14 – 1.18)	0.71 (0.13 - 1.26)	0.74 (0.10 - 1.51)	0.47 (0.04 - 1.28)	Not more than 0.5 mg/l for annual mean
Total Nitrogen (TN) (mg/L)	0.59 (0.36 - 0.86)	0.78 (0.33 - 1.32)	0.81 (0.33 - 1.40)	0.89 (0.31 - 1.48)	0.93 (0.25 - 1.72)	0.63 (0.21 - 1.47)	-
Orthophosphate Phosphorus (PO_4) (mg/L)	0.021 (0.010 - 0.032)	0.023 (0.008 - 0.041)	0.025 (0.008 - 0.043)	0.028 (0.008 - 0.050)	0.022 (0.006 - 0.041)	0.015 (0.006 - 0.030)	-
Total Phosphorus (TP) (mg/L)	0.04 (<0.02 - 0.07)	0.04 (<0.02 - 0.05)	0.04 (<0.02 - 0.06)	0.05 (<0.02 - 0.06)	0.04 (<0.02 - 0.07)	0.03 (<0.02 - 0.06)	-
Chlorophyll-*a* (µg/L)	2.0 (0.4 - 6.4)	2.3 (0.3 - 7.1)	2.7 (0.3 - 11.0)	3.3 (0.3 - 14.3)	3.8 (0.5 - 13.7)	3.8 (0.3 - 12.3)	-
E. coli (cfu/100 mL)	300 (36 - 2200)	71 (7 - 850)	110 (8 - 1100)	190 (10 - 4100)	22 (4 - 180)	3 (<1 - 11)	-

Note:

(1) Data source: Marine Water Quality in Hong Kong in 2014 (EPD).
(2) Data presented are annual arithmetic means of depth-averaged results except for *E.coli* and faecal coliforms that are annual geometric means.
(3) Data in brackets indicate the ranges.

5.9 Assessment Approach and Methodology

5.9.1 The assessment area includes all areas within 300 m from the Project boundary, and covers relevant water sensitive receivers (WSRs) that have a bearing on the environmental acceptability of the Project within the Deep Bay and North Western WCZ. The assessment area should be extended to include other areas such as stream courses and associated water systems, existing and planned drainage system if they are found being impacted during the course of the EIA study and have a bearing on the environmental acceptability of the Project.

5.9.2 The existing information regarding the assessment area have been reviewed. Relevant information sources include:

- EPD River Water Quality in Hong Kong in 2014;
- EPD Marine Water Quality in Hong Kong in 2014;
- Government aerial photos;
- GeoInfo Map from Lands Department;
- Approved EIA Report for Deep Bay Link;
- Approved EIA Report for Shenzhen Western Corridor;
- Approved EIA Report for Upgrading and Expansion of San Wai Sewage Treatment Works (SW STW) and Expansion of Ha Tsuen Sewage Pumping Station;
- Approved EIA for Hang Hau Tsuen Channel at Lau Fau Shan; and
- Approved EIA for Expansion of Hong Kong International Airport into a Three-Runway System.

5.9.3 The WSRs that may be affected by the Project have been identified. Water sampling and laboratory testing were conducted to quantify the water quality of the WSRs. Potential sources of water quality impact that may arise during the construction works and operational activities were described. This task included identifying pollutants from point discharges and non-point sources that could affect the quality of surface water run-off. All the identified sources of potential water quality impact were then evaluated and their impact significance determined. The need for mitigation measures to reduce any identified adverse impacts on water quality to acceptable levels was determined.

5.10 Identification and Evaluation of Environmental Impacts – Construction Phase

5.10.1 No marine works would be required for construction of the Project. Only inland construction works would be involved under the Project including:

- Site formation and demolition of existing buildings;
- Construction of new infrastructure including: footbridges, cycle bridges, new buildings, Flushing Water Service Reservoirs (FLWSR), Fresh Water Service Reservoirs (FWSR), sewerage / drainage / road / street networks, SPS, new HSK STW, eco trail, cultural heritage trail, cycle track / cycle way, parks, Environmentally Friendly Transport Services (EFTS), EFTS depot, proposed West Rail HSK Station (HSK Station), Public Transport Interchange (PTI), petrol / Liquefied Petroleum Gas (LPG) filling stations, Refuse Transfer Station (RTS), District Cooling System (DCS), electricity substations, channel revitalisation and greening works and flood retention facilities (involving flood retention lakes), etc.;

5.10.2 The potential sources of water quality impacts arising from the inland construction works of the Project include:

- General construction activities;
- Construction site run-off;
- Construction works near watercourses;
- Construction works in watercourses / reinforced concrete flood storage pond;
- Removal or diversion of watercourses;
- Removal or filling of ponds and wet areas;
- Accidental spillage;
- Sewage from construction workforce;
- Groundwater from contaminated areas, contaminated site run-off and wastewater from land decontamination; and
- Sewerage diversion works.

5.10.3 Construction of DP1, DP2, DP5, DP6, DP12 (which are road projects) and DP9 (which involves SPS) as mentioned in **Section 5.1.2** above would also involve land-based construction activities. The potential sources of water quality impacts arising from DP1, DP2, DP5, DP6, DP9 and DP12 during the construction phase would be the same as that identified for the Project as listed in **Section 5.10.2** above.

5.10.4 Assessments of the potential water quality impacts arising from the construction of the Project including DP1, DP2, DP5, DP6, DP9 and DP12 are provided in **Sections 5.10.5 to 5.10.43** below.

General Construction Activities

5.10.5 Effluent discharged from temporary site facilities should be controlled to prevent direct discharge to the neighbouring inland waters and stormwater drains. Such effluent may include wastewater resulting from dewatering processes and wheel washing of site vehicles at site entrances. Discharge of debris and rubbish such as packaging, construction materials and refuse would also be the potential water pollution sources. Adoption of the guidelines and good site practices for handling and disposal of construction discharges as part of the construction site management practices (as given in **Sections 5.13.1 to 5.13.14**) would minimise the potential impacts.

Construction Site Run-off

5.10.6 The Project development is divided by stages as shown in **Figure 2.18**. The average daily run-off generated from the construction sites for different phases of the Project development is given in **Table 5.12** below. The total works area for Stage 3 development (of approximately 2 km^2) would be amongst the largest, with an estimated daily run-off of about 1,144 m^3. The construction site run-off generated from all other stages of the Project development (with smaller works areas) would be smaller. The assumptions adopted for quantification of the daily construction site run-off are given in **Appendix 5.1**.

Table 5.12 Averaged Daily Construction Site Run-off for the Project by Stages

Phase (refer to Figure 2.18)	Approximate Area (km^2)	Approximate Daily Volume of Construction Site Run-off (m^3/day)
ASEP Phase 1 (see Note 1)	0.002	1
ASEP Phase 2 (see Note 1)	0.038	22
ASEP Phase 3 (See Note 1)	0.470	268
Stage 1	0.024	14
Stage 2	1.365	777
Stage 3	2.008	1,144
Stage 4	0.619	353

Note 1: ASEP – Advance Site Formation and Engineering Infrastructure Project

5.10.7 Using the same approach and assumptions as given in **Appendix 5.1**, the average daily run-off generated from the construction sites of DP1, DP2, DP5, DP6 and DP12 (which are road projects) and DP9 (which are SPS) would range from about 2 to 175 m^3 as summarised in **Table 5.13** below.

Table 5.13 Averaged Daily Construction Site Run-off for DP1, DP2, DP5, DP6, DP9 and DP 12

DP (refer to Section 5.1.2)	Approximate Area (km^2)	Approximate Daily Volume of Construction Site Run-off (m^3/day)
DP1	0.108	62
DP2	0.307	175
DP5	0.009	5
DP6	0.005	3
DP9	0.009	5
DP12	0.004	2

5.10.8 As compared to the design flow capacity of the new HSK STW of 85,500 m^3/day as mentioned in **Section 5.11.3** below, the quantity of construction site run-off estimated for the Project is considered small. It should be highlighted that the daily construction site run-off (estimated using average rainfall) provided above is representative of an average or typical condition. The actual construction site run-off during the construction period would vary, i.e. could be higher during wet season (e.g. April to October) and lower in dry season (e.g. November to March).

5.10.9 Potential pollution sources of site run-off may include:

- Run-off and erosion of exposed bare soil and earth, drainage channel, earth working area and stockpiles;
- Release of any bentonite slurries, concrete washings and other grouting materials with construction run-off or stormwater;
- Wash water from dust suppression sprays and wheel washing facilities; and
- Fuel, oil and lubricants from maintenance of construction vehicles and equipment.

5.10.10 During rainstorms, site run-off would wash away the soil particles on unpaved lands and areas with the topsoil exposed. The run-off is generally characterised by high concentrations of SS. Release of uncontrolled site run-off would increase the SS levels and turbidity in the nearby water environment. Site run-off may also wash away contaminated soil particles and therefore cause water pollution.

5.10.11 Wind blown dust would be generated from exposed soil surfaces in the works areas. It is possible that wind blown dust would fall directly onto the nearby water bodies when a strong wind occurs. Dispersion of dust within the works areas may increase the SS levels in surface run-off causing a potential impact to the nearby sensitive receivers.

5.10.12 It is important that proper site practice and good site management be followed to prevent run-off with high level of SS from entering the surrounding waters. Best Management Practices (BMPs) in controlling construction site discharges are recommended for this Project as described in **Section 5.13.1 to 5.13.14**. In view of the sensitive nature of the Deep Bay WCZ, an Emergency Response Plan (ERP) should also be developed to minimise the potential impact from construction site discharges under failure of treatment facilities during emergency situations or inclement weather. An outline of the ERP is provided in **Sections 5.13.34 to 5.13.35**. With the implementation of BMPs to control construction site effluents and ERP to minimise water quality impact under emergency situations, disturbance of water bodies and deterioration in water quality would be minimized.

Construction Works near Watercourses

5.10.13 Watercourses are located within the Project area as identified in **Section 5.3**. Construction works near these watercourses may pollute the stormwater or inland waters due to potential release of construction wastes. Construction wastes are generally characterised by high concentration of SS and elevated pH. Adoption of good housekeeping and mitigation measures would reduce the generation of construction wastes and potential water pollution. The implementation of measures to control run-off and drainage water will be important for the construction works adjacent to the inland water in order to prevent run-off and drainage water with high levels of SS from entering the water environment. With the implementation of adequate construction site drainage and BMPs as described in **Sections 5.13.1 to 5.13.14** and provision of mitigation measures as specified in ETWB TC (Works) No. 5/2005 “Protection of natural streams/rivers from adverse impacts arising from construction works” as detailed in **Section 5.13.15**, it is anticipated that water quality impacts would be minimised.

5.10.14 Sections of existing drainage channels including TSW Main Channel, HSK Channel and Tin Sam Channel, running through the Project will be revitalised (e.g. enhancing the greening of the channel banks through the provision of “grasscrete” or similar products). The “grasscrete” or similar products will only be placed on the channel banks above the channel water and the works will be conducted during the dry season when the water flow is low. No underwater works would be required for the revitalisation works. The proposed revitalisation works would involve handling of concrete and “grasscrete” which contain negligible fines content, and as such no loss of fines into the watercourses would be expected. The implementation of measures to control run-off and drainage water will be important for the construction works adjacent to the revitalised drainage channel water in order to prevent run-off and drainage water with high levels of SS from entering the revitalised drainage channel. With the implementation of adequate construction site drainage and BMPs as described in **Sections 5.13.1 to 5.13.14** and provision of mitigation measures as specified in ETWB TC (Works) No. 5/2005 “Protection of natural streams/rivers from adverse impacts arising from construction works” as detailed in **Section 5.13.16**, no unacceptable water quality impacts would be anticipated.

Construction Works in Watercourses / Concrete Flood Storage Pond

5.10.15 Two small footbridges and two small cycle bridges proposed under the Project would run across the existing watercourses. The two footbridges would run across the TSW Main Channel near Sites 1-4 and 1-8 respectively. The two cycle bridges would run across the upstream tributary of TSW Main Channel near the western and southern boundary of Site 1-27 respectively. The locations of these footbridges and cycle bridges are shown in **Figure 5.6**.

5.10.16 All the piers of these footbridges and cycle bridges would be land-based and located outside the watercourses during the operational phase. However, as these piers are located close to the bank of the watercourses, construction of these minor bridge piers would involve temporary construction activities (e.g. piling and excavation) in the watercourses.

5.10.17 A north bound slip road from KSWH is proposed to provide direct access via Road D6 to west part of the Project which will be located next to KSWH. The slip road and part of Road P1 will be partially located in the "Conservation Area" designation of Yuen Tau Shan. The alignment of Road P1 and the slip road will be constructed within an existing reinforced concrete flood storage pond to avoid loss of natural/semi-natural habitats. Construction of the road alignment in the concrete flood storage pond would involve installation of bored piles, construction of pile caps and piers for the proposed viaducts. Locations of the foundation and pile / pier construction works in the reinforced concrete flood storage pond are indicatively shown in **Figure 5.7**.

5.10.18 Potential impacts may be generated by discharge of construction materials, wastewater, sediment and spillage to the receiving waters in the downstream. The construction method and sequence of the proposed construction in watercourses / pond should be carefully designed so that all the construction works including any excavation and piling operations would be undertaken within a dry zone and physically separated from the water flow in the existing watercourses / pond.

5.10.19 Impermeable sheet pile walls or cofferdam walls or steel casing would be installed to fully enclosed the construction works area or the piling works in watercourse / pond prior to the commencement of any works in watercourse / pond. Dewatering of the construction works area or diversion of water flow would be undertaken before the construction works to avoid water flow in the construction works area. Silt removal facilities would be used to clarify the effluent generated from the dewatering operation before discharging back to the watercourse / drainage system as a precautionary measure. Any construction works including excavation and piling activities would be undertaken in a dry zone surrounded by the impermeable sheet pile walls or cofferdam walls or steel casing. Depending on the water level in the watercourses / pond during the construction, where practicable, silt curtain would be installed around the construction works area inside the watercourse / pond as a second layer of protection to further minimise sediment and contaminant release. All wastewater generated from the piling activities would be considered as part of the construction site effluent, which would be properly collected and treated as appropriate to meet the standards stipulated in the TM-DSS before disposal.

5.10.20 As the scale of the bridge piers is minor, the temporary works areas in the watercourses would be small and thus no significant obstruction of the water flow would be expected from the pier construction works.

5.10.21 The existing concrete flood storage pond is only used to retain some stormwater during heavy rains and it is usually dry in normal days and thus no impact in relation to hydrodynamics would be expected from the foundation / pile / pier construction in the pond.

5.10.22 Excavated materials (including sediment) may be generated from the construction works. Mitigation measures for handling and disposal of excavated materials and sediment as recommended in **Section 5.13.37** should be followed to minimise the potential environmental impacts arising from the excavated materials.

5.10.23 It is also recommended that the construction works in watercourses / pond should be undertaken in dry season, where practicable, when the water flow is low. With the adoption of the construction method as described above, together with the adequate construction site drainage as recommended in **Sections 5.13.1 to 5.13.14** as well as the protection measures for construction works near watercourses as detailed in **Section 5.13.15**, it is anticipated that water quality impacts would be minimised.

Removal / Diversion of Watercourses

5.10.24 The TSW Main Channel and its tributaries (TSW) consist of a core channel (namely TSW Main Channel), three major tributaries streams (namely Lo Uk Tsuen Tributaries, Tin Sam Channel and HSK Main Street Channel respectively) and other minor tributaries, which spread across the majority of the Project area as shown in **Figure 5.1**. The TSW Main Channel and majority of its upstream tributaries will be retained under the Project. Only several minor upstream watercourses will be removed under the Project. These minor watercourses include: a minor upstream watercourse (namely TSW3) of Lo Uk Tsuen Tributaries; four minor upstream sections (namely TSW1A, TSW1B, TSW1C and TSW1D respectively) that finally join the Tin Sam Channel; and four other minor watercourses (namely TSW2, TSW2A, TSW2B and TSW5 respectively) that finally drain to the TSW Main Channel as summarised in **Table 5.14** below. In addition, a section (namely TSW4) of the Tin Sam Channel will be realigned under the Project.

5.10.25 Hang Hau Tsuen Channel and its tributaries (HHT) are located in the northern tip of the Project area as shown in **Figure 5.1**. A side branch (namely HHT1) and the most upstream part (namely HH2) of the Hang Hau Tsuen Channel will be removed under the Project.

5.10.26 The upstream tributary of Tuen Mun River (TMR) is located in the southern tip of the Project area as shown in **Figure 5.1**. The most upstream part (namely TMR1) of this tributary will be removed under the Project.

5.10.27 A summary of the minor watercourses to be removed or diverted under this Project is provided in **Table 5.14** below.

Table 5.14 Minor Sections of Watercourses to be Removed or Diverted

Name of Water Section to be Removed or Diverted	Description	Remarks	Figure Reference
Water Sections belong to the tributaries of Tin Shui Wai Main Channel (TSW):			
TSW3	A small upstream water section of Lo Uk Tsuen Tributaries (in the middle west portion of the Project area)	To be removed under the Project	Figure 5.2.8 (see middle left)
TSW1A	A minor water section (in the southern Project area) that finally drain into Tin Sam Channel	To be removed under the Project	Figure 5.2.9 (see left hand side)
TSW1B	A minor water section (in the southern Project area) that finally drain into Tin Sam Channel	To be removed under the Project	Figure 5.2.9 (see left hand side)

Name of Water Section to be Removed or Diverted	Description	Remarks	Figure Reference
TSW1C	A minor water section (in the southern Project area) that finally drain into Tin Sam Channel	To be removed under the Project	Figure 5.2.9 (see left hand side)
TSW1D	A minor water section (in the southern Project area) that finally drain into Tin Sam Channel	To be removed under the Project	Figure 5.2.9 (see left hand side)
TSW4	A section of Tin Sam Channel (in the southern Project area)	To be realigned under the Project	Figure 5.2.9 (see left hand side)
TSW2	A minor water section (in the northern Project area) that finally drain into the TSW Main Channel	To be removed under the Project	Figure 5.2.7 (see upper right)
TSW2A	A minor water section (in the northern Project area) that finally drain into the TSW Main Channel	To be removed under the Project	Figure 5.2.7 (see upper right)
TSW2B	A minor water section (in the northern Project area) that finally drain into the TSW Main Channel	To be removed under the Project	Figure 5.2.7 (see upper right)
TSW5	A minor water section (in the eastern tip of Project area) that finally drain into the TSW Main Channel	To be removed under the Project	Figures 5.2.12 (see lower left), Figure 5.2.13 (see upper left corner)
Water Sections belong to the tributaries of Hang Hau Tsuen (HHT):			
HHT1	A side branch Hang Hau Tsuen Channel in the northern Project area	To be removed under the Project	Figure 5.2.1
HHT2	An upstream section of Hang Hau Tsuen Channel in the northern Project area	To be removed under the Project	Figure 5.2.7 (see upper middle)
Water Section belongs to the tributaries of Tuen Mun River (TMR):			
TMR1	An upstream section of the Tuen Mun River in the southern tip of Project area	To be removed under the Project	Figure 5.2.5 (see upper right)

5.10.28 Removal of existing watercourses (namely TSW1A, TSW1B, TSW1C, TSW1D, TSW2 TSW2A, TSW2B, TSW3, TSW5, HHT1, HHT2, TMR1) would involve diversion of water flow from their existing routes to the new routes through the proposed covered drainage system of the new development and demolition of the existing watercourse. Only a section (namely TSW4) of Tin Sam Tsuen would be diverted to a new open channel nearby. Construction of the new channel and new drainage network may involve excavation and use of concrete. Potential impacts may be generated by discharge of concrete slurry and other grouting materials generated by concreting works as well as the release of construction materials, wastewater, excavated sediment, spillage and contaminants to the receiving waters in the downstream (due to soil excavation for construction of new drainage and demolition of the existing watercourses). All these construction works should be undertaken in dry conditions to avoid potential water quality impacts upon the downstream water quality. The tentative works sequence for diversion and removal of watercourse is described below.

5.10.29 Construction works at watercourse would be undertaken only after flow diversion or dewatering operation is fully completed to avoid water flow in the works area. Dewatering of watercourse would be performed by diverting the water flow to new or temporary drainage. Where necessary, cofferdams or similar impermeable sheet pile walls should be used to isolate the works areas from neighbouring waters. The permanent or temporary drainage for carrying the diverted flow from existing watercourse to be removed would be constructed and completed before dewatering of that existing watercourse. Construction of all the proposed permanent and temporary drainage would be undertaken in a dry condition prior to receiving any water flow.

5.10.30 Excavated materials (including sediment) may be generated from the removal and diversion of watercourses. Mitigation measures for handling and disposal of excavated materials and sediment as recommended in **Section 5.13.37** should be followed to minimise the potential environmental impacts arising from the excavated materials.

5.10.31 Mitigation measures for protection of downstream water quality from diversion and removal of watercourses are described in **Sections 5.13.20 to 5.13.23**. With adoption of the recommended mitigation measures, there should not be significant sediment and contaminant release to the downstream water.

Removal / Filling of Ponds and Wet Areas

5.10.32 Due to the new developments, three existing ponds (namely P8B, P8C and P10 respectively as shown in **Figures 5.2.7** and **5.2.9**) and one wet agricultural area (namely A3A as shown in **Figures 5.2.5**) within the Project area will be completely removed under this Project. In addition, part of all the three wet agricultural areas (namely A2B, A2C and A2D respectively as shown in **Figure 5.2.8**) located outside the existing "V" area will be removed under the Project. **Table 5.15** below summarises all the ponds and wet agricultural areas to be completely or partially removed under the Project.

Table 5.15 Ponds and Wet Areas to be Completely or Partially Removed

ID	Description	Location	Within Project Area? (Yes /No)	Figure Reference	Remarks
P8B	Inactive fish pond	At the northern part of Project area to the south of Fung Kong Tsuen	Yes	Figure 5.2.7	P8B will be removed under the Project
P8C	Inactive fish pond	At the northern part of Project area to the south of Fung Kong Tsuen	Yes	Figure 5.2.7	P8C will be removed under the Project
P10	Inactive fish pond	At the southern portion of Project area to the west of San Lee Uk Tsuen	Yes	Figure 5.2.9	P10 will be removed under the Project
A2B	Wet agricultural area	At the border of "V" area in middle portion of the Project area	Yes	Figure 5.2.8	Part of A2B, A3C, A3D located outside the "V" will be removed under the Project
A2C	Wet agricultural area	At the border of "V" area in middle portion of the Project area	Yes	Figure 5.2.8	
A2D	Wet agricultural area	At the border of "V" area in middle portion of the Project area	Yes	Figure 5.2.8	
A3A	Wet agricultural area	At the southern part of the Project area near the upstream tributaries of Tuen Mun River	Yes	Figure 5.2.5	A3A will be removed under the Project

5.10.33 All the ponds and wet areas to be removed are isolated and not connected to any existing watercourse. The associated construction works would include draining the water in ponds / watercourses before filling up these areas or before commencement of any excavation and construction works. The water of these ponds and wet areas to be drained would probably be sediment-laden and would carry a certain level of pollutants. Direct dumping of these drained waters to the nearby watercourse or marine water will not be allowed.

5.10.34 The drained water generated from dewatering of the ponds / wet areas to be removed should be temporarily stored in appropriate storage tanks or containers for reuse on-site as far as practicable and any surplus water should be tankered away for disposal at the STW. Any surplus water should be treated as necessary before disposal to the STW in compliance with the TM-DSS. In order to minimise the potential impact, dewatering works at ponds / wet areas should be conducted within dry season to minimise the quantity of drained water. No direct discharge of drained water from these construction works will be allowed.

5.10.35 Excavated materials (including sediment) may be generated from the construction works in ponds and wet areas. Mitigation measures for handling and disposal of excavated materials and sediment as recommended in **Section 5.13.37** should be followed to minimise the potential environmental impacts arising from the excavated material. Together with the adoption of the mitigation measures for removal of ponds and wet areas as recommended in **Sections 5.13.24 and 5.13.25**, no unacceptable water quality impact would be expected.

Accidental Spillage

5.10.36 The use of engine oil and lubricants, and their storage as waste materials has the potential to create impacts on the water quality if spillage occurs and enters adjacent water environment. Waste oil may infiltrate into the surface soil layer, or run-off into marine water environment, increasing hydrocarbon levels. The potential impacts could however be mitigated by practical mitigation measures and good site practices (as given in **Sections 5.13.26 to 5.13.28**).

Sewage from Construction Workforce

5.10.37 Sewage effluents will arise from the sanitary facilities provided for the on-site construction workforce. Based on the DSD Sewerage Manual, the sewage production rate for construction workers is estimated at 0.35 m^3 per worker per day. Thus, for every 100 construction workers working simultaneously at the construction site, about 35 m^3 of sewage would be generated per day. The characteristics of sewage would include high levels of BOD_5, Ammonia and *E. coli* counts.

5.10.38 Portable chemical toilets would be provided for handling the construction sewage generated by the workforce. The number of the chemical toilets required for the construction sites would be subject to later detailed design, the capacity of the chemical toilets, and contractor's site practices. A licensed contractor would be employed to provide appropriate and adequate portable toilets and be responsible for appropriate disposal and maintenance.

5.10.39 Notices would be posted at conspicuous locations to remind the workers not to discharge any sewage or wastewater into the nearby environment during the construction phase of the Project. Regular environmental audit on the construction site would be conducted in order to provide an effective control of any malpractices and achieve continual improvement of environmental performance on site.

5.10.40 Provided that sewage is not discharged directly into stormwater drains or inland waters, and temporary sanitary facilities are used and properly maintained, no adverse water quality impact would be anticipated. Mitigation measures and good site practices as given in **Sections 5.13.29 and 5.13.30** should be implemented.

Groundwater from Contaminated Areas, Contaminated Site Run-off and Wastewater from Land Decontamination

5.10.41 It is identified that some of the construction works areas would have land contamination issues. Proper land contamination remediation and mitigation measures are proposed in **Section 8**. Any contaminated material disturbed, or material which comes into contact with the contaminated material, has the potential to be washed with site run-off into watercourses. Any wastewater discharge from land decontamination processes could also adversely affect the nearby water environment. Excavated contaminated materials would be properly stored, housed and covered to avoid generation of contaminated run-off. Open stockpiling of contaminated materials will not be allowed. Any contaminated site run-off and wastewater from land decontamination activities will be properly treated and disposed in compliance with the requirements of the TM-DSS. Mitigation measures for contaminated site run-off and wastewater from land decontamination are recommended in **Section 5.13.31**. With proper implementation of the recommended mitigation measures, the potential water quality impacts arising from the land decontamination works would be minimised.

5.10.42 Groundwater pumped out or from dewatering process during excavation works in the contaminated areas would be potentially contaminated. Any contaminated groundwater will be either properly treated or properly recharged into the ground in compliance with the requirements of the TM-DSS. No direct discharge of contaminated groundwater will be adopted. Mitigation measures and monitoring requirements for contaminated groundwater discharge / recharge are recommended in **Sections 5.13.32 to 5.13.33**. With proper implementation of the recommended mitigation measures, no unacceptable water quality would be expected from the groundwater generated from contamination areas.

Sewerage Diversion

5.10.43 As discussed in **Section 6.8**, construction of the Project would involve diversion of the existing twin 800 mm diameter rising mains along Tin Ying Road. New sewerage facilities for receiving the diverted sewage flow from the existing rising mains will be constructed prior to the commencement of any demolition and construction works at the existing rising mains. All sewage flow running in the existing rising mains along Tin Ying Road will be diverted to the new sewerage system prior to any demolition and construction works at the rising mains. No discharge of sewage flow to the environment will be allowed during the sewerage diversion works and hence, no unacceptable water quality would result.

5.11 Identification and Evaluation of Environmental Impacts – Operational Phase

Sewage Disposal Strategy for the New Developments

5.11.1 Sewage effluent will be generated from the new developments. A public sewerage system will be built to collect and convey all the sewage effluents generated from the Project area to STW for proper disposal.

5.11.2 The existing SW STW is a preliminary treatment plant located within the Project area with a flow capacity of 164,000 m^3/day. Under the approved EIA study for "Upgrading and expansion of SW STW and expansion of Ha Tsuen Sewage Pumping Station (SW STW)" completed in 2003, it was proposed to expand the design flow of SW STW to 246,000 m^3/day and upgrade its treatment level to Chemically Enhanced Primary Treatment (CEPT) plus UV disinfection. This approved EIA concluded that the proposed upgrading works would be environmentally acceptable. However, during the later detailed design stage under Agreement No. CE43/2007 (DS) (3), the capacity of SW STW (Phase 1) has been revised to 200,000 m^3/day. The treatment level of SW STW would also be upgraded to CEPT plus UV disinfection under Agreement No. CE43/2007 (DS). According to the tentative construction programme of Contract No. DC/2013/10, the upgrading of San Wai STW Phase 1 would be completed in 2020.

5.11.3 With reference to the assessment Sewerage and Sewage Treatment Implications in **Chapter 6**, a new Hung Shui Kiu Sewage Treatment Works (HSK STW) will be built by phases within the Project area to handle the sewage generated from the Project (with reference to **Section 6.6**). Phase 1 of HSK STW would be commissioned in 2031 to provide a flow capacity of 70,000 m^3/day. Phase 2 of HSK STW would involve further expansion of the total capacity of HSK STW to 85,500 m^3/day for commissioning in 2037/38 subject to further review on latest population data and sewage in San Wai Sewage Catchment. The Treated Sewage Effluent (TSE) from the new HSK STW would be pumped into the North West New Territories Effluent Tunnel (hereafter referred to as the "NWNT Tunnel") and discharged at the Urmston Road Outfall located in the marine water of the North Western WCZ. The indicative location of the Urmston Road Outfall is shown in **Figure 5.4**.

5.11.4 In addition, a new Yuen Long South (YLS) STW with a planned design capacity of 23,000 m^3/day is proposed under a separate study (namely "Agreement No. CE 35/2012 (CE) Planning and Engineering Study for Housing Sites in Yuen Long South – Investigation") to handle the sewage flow generated from the Potential Development Area (PDA) in YLS. Various TSE disposal options are being considered for the new YLS STW under separate study including: (1) pumping the TSE into the NWNT Tunnel and then discharged at the Urmston Road Outfall, which would have potential cumulative impact upon the marine water of North Western WCZ together with the TSE discharge from HSK STW and SW STW; and (2) local discharge of TSE in Deep Bay WCZ. Under this EIA study, only the local disposal option of TSE from the proposed YLS STW in the Deep Bay WCZ would be considered. It should be noted that the final TSE disposal scheme for the proposed YLS STW is not confirmed at the time of this EIA study and subject to further studies.

5.11.5 Installation of water reclamation facilities is proposed under this Project to reduce the TSE discharge. The water reclamation facilities would provide tertiary treatment for some of the TSE from the HSK STW for production of reclaimed water for beneficial use (e.g. toilet flushing). Details of the proposed water reclamation facility for this Project are provided in **Sections 5.11.52 to 5.11.55**. Based on the best available information obtained from this Project, with implementation of the proposed water reclamation facilities, the total TSE discharge from HSK STW (Phase 1) and HSK STW (Phase 1 and Phase 2) could be reduced to about 31,051 m^3/day and 45,696 m^3/day, respectively. The water reclamation facilities including the FLWSR proposed under this Project are tentatively scheduled for commissioning before 2031. Details of the sewage flow projections, design of the proposed water reclamation facilities, sewerage and STW for the Project are provided under the Sewerage and Sewage Treatment Implications in **Chapter 6**.

(3) Yuen Long and Kam Tin sewerage treatment upgrade - Upgrading of San Wai Sewage Treatment Works - Investigation, Design and Construction

5.11.6 Since all the sewage effluent generated from the new developments of this Project will be properly treated and discharged to the North Western WCZ. This Project will not induce any net increase in sewage loading to the Deep Bay waters under normal plant operation. The potential water quality impact due to local disposal of TSE from YLS STW in the Deep Bay WCZ will be assessed under separate detailed EIA study for YLS STW.

5.11.7 The planning and engineering study of this Project is a Schedule 3 Designated Project (DP) under the EIAO, whilst there will be a Schedule 2 DP for the new HSK STW. The design of the new HSK STW (such as the plant layout) is not available at this Schedule 3 EIA Stage. The detailed water quality impact of the new HSK STW will be further investigated in a separate Schedule 2 EIA under the EIAO based on the updated and latest information on the treatment plant design and sewage flow projections for this Project and other concurrent projects in the assessment area. The reclaimed water quantities adopted in this EIA are preliminary in nature and subject to review under separate further studies. It is recommended that opportunities should be explored to maximise the use of reclaimed water and reduce the TSE discharge as far as possible under the detailed EIA studies for HSK STW.

Pollution Loading to North Western WCZ

5.11.8 As mentioned in **Sections 5.11.1 to 5.11.5** above, the existing and likely future pollution loading to North Western WCZ would include the point source discharges from the Urmston Road Outfall. **Table 5.16** below gives an estimation on the flow and loading of TSE discharged from the Urmston Road Outfall under the existing and likely future conditions. Under the existing condition, the pollution loading was estimated based on the design capacity (164,000 m^3/day) of the existing SW STW, which is currently a preliminary treatment works with discharges to the Urmston Road Outfall. Under the future condition with implementation of this Project as discussed in **Sections 5.11.1 to 5.11.5** above, the Urmston Road Outfall would potentially receive TSE flow from the upgraded SW STW and the new HSK STW. It is assumed that the future upgraded SW STW proposed under Agreement no. CE43/2007(DS) would provide CEPT plus UV disinfection, whilst the new HSK STW would employ secondary treatment plus UV disinfection and 75% nitrogen (N) removal as mentioned in **Sections 6.6 and 6.11**. The effluent concentrations for different treatment processes used for compiling the TSE flow and load are given in **Appendix 5.2**. As mentioned in **Section 5.11.2** above, the TSE discharge (after CEPT and disinfection) from the future upgraded SW STW was assumed to be 246,000 m^3/day under the approved EIA for SW STW, which concluded that such a discharge to North Western WCZ would be environmentally acceptable. As such, the point source loading to North Western WCZ assumed in the approved EIA for SW STW is also included in **Table 5.16** below for comparison.

Table 5.16 Existing and Likely Future Point Source Loading to North Western WCZ from Urmston Road Outfall

Parameter	Unit	Existing Condition, see Note 1	Likely Future Condition Assumed in Approved EIA for SW STW (without this Project), see Note 1	Likely Future Condition Assumed under this EIA (with this Project) see Notes 1 & 2		
				SW STW Phase 1 only (2026)	plus HSK STW Phase 1 (2031)	plus HSK STW Phase 2 (2037/38)
TSE Discharge	m^3 /day	164,000 (see Note 4)	246,000 (see Note 5)	200,000 (see Note 6)	231,051 (see Note 7)	245,696 (see Note 8)
Biochemical Oxygen Demand (BOD)	kg/day	37,064	24,600	20,000	20,621	20,914
Suspended Solids (SS)	kg/day	33,784	13,530	11,000	11,932	12,371

Parameter	Unit	Existing Condition, see Note 1	Likely Future Condition Assumed in Approved EIA for SW STW (without this Project), see Note 1	Likely Future Condition Assumed under this EIA (with this Project) see Notes 1 & 2		
				SW STW Phase 1 only (2026)	plus HSK STW Phase 1 (2031)	plus HSK STW Phase 2 (2037/38)
Ammonia Nitrogen (NH_3-N)	kg/day	3,570	6,150	5,000	5,062	5,091
Organic Nitrogen (Org-N)	kg/day	2,952	2,165	1,760	1,822	1,851
Total Inorganic Nitrogen (TIN), see Note 3	kg/day	3,928	6,150	5,000	5,248	5,366
Total Nitrogen (TN)	kg/day	6,880	8,315	6,760	7,071	7,217
Total Phosphorus (TP)	kg/day	940	556	452	522	555
E. coli	no./day	2.89E+16	4.92E+08	4.00E+08	4.03E+08	4.05E+08

Note:

(1) The effluent concentrations / standards used for compiling the pollution loading are detailed in Appendix 5.2. The existing and upgraded SW STW would adopt preliminary treatment and CEPT with disinfection respectively. The HSK STW (Phase 1 and Phase 2) is assumed to employ secondary treatment plus disinfection and 75% N removal subject to further detailed EIA studies. The effluent standards of the HSK STW is subject to further studies during the EIA stage of this STW.

(2) TSE discharge from YLS STW is assumed to be disposed locally in Deep Bay WCZ under this EIA study.

(3) Total inorganic nitrogen (TIN) is equal to the sum of ammonia nitrogen (NH3-N), nitrate nitrogen (NO3-N) and nitrite nitrogen (NO2-N)

(4) Design flow of existing SW STW (with preliminary treatment)

(5) Design flow of upgraded SW STW assumed in the approved EIA for SW STW

(6) Design flow of upgraded SW STW (Phase 1). The flow due to the new development of HSK NDA in year 2026 (14,538 m3/day as mentioned in Table 6.9) is assumed to be treated by SW STW (see Section 6.6.3).

(7) noCombined TSE flow of upgraded SW STW and new HSK STW (Phase 1) assumed in this EIA, with consideration of the flow reduction due to the proposed water reclamation facilities. According to Section 5.11.5 and Table 6.9, about 31,051 m3/day will be treated by secondary plus treatment (with UV disinfection and 75% nitrogen removal) and discharged to Urmston Road Outfall during HSK STW , while the rest (45,137 m3/day - 31,051 m3/day = 14,086 m3/day) will be treated by tertiary treatment for reclaimed water reused in the Project.

(8) Combined TSE flow of upgraded SW STW and new HSK STW (Phase 1 and 2) assumed in this EIA, with consideration of the flow reduction due to the proposed water reclamation facilities. According to Section 5.11.5 and Table 6.10, about 45,696 m3/day will be treated by secondary plus treatment (with UV disinfection and 75% nitrogen removal) and discharged to Urmston Road Outfall from HSK STW , while the rest (85,500 m3/day - 45,696 m3/day = 39,804 m3/day) will be treated by tertiary treatment for reclaimed water reused in the Project.

5.11.9 As indicated in the table above, the future pollution loading from upgraded SW STW (with CEPT and disinfection) assumed in the approved EIA for SW STW would be smaller than that of the existing discharge from SW STW (with preliminary treatment only) for all the selected parameters except for inorganic nitrogen (i.e. NH_3-N). This is due to the low NH_3-N removal efficiency of the CEPT process in combination with the significant increase in the design capacity to 246,000 m^3/day in the future. The CEPT removal efficiencies for other parameters are much greater, which counter the effect of the flow increase. Under all the future scenarios and with reference to the final design adopted under Agreement no. CE43/2007(DS), the flow capacity of upgraded SW STW (with CEPT and disinfection) would be reduced, whilst more advance treatment (i.e. secondary treatment plus disinfection and 75% N removal) is assumed for the new HSK STW as discussed in **Section 5.11.8** above. As a result, the overall flow and loading discharged at the Urmston Road Outfall estimated under this EIA would be smaller than that adopted in the approved EIA for SW STW for all selected parameters. Hence, the associated water quality impact upon the marine water of North Western WCZ would not be worse than that predicted in the approved EIA for SW STW. There will be no increase in both the flow and load as compared to the condition assumed in the approved EIA for SW STW. Thus, no adverse changes in hydrology, flow regime and water quality in the receiving marine water of North Western WCZ resulted from the Project as compared to

the water quality prediction in the approved EIA for SW STW. Opportunities to maximise the use of reclaimed water would be further explored to minimise the TSE flow at the Urmston Road Outfall during the detailed EIA studies for HSK STW. The potential water quality impact due to the TSE discharge from the new HSK STW will be assessed and ascertained in separate EIA studies.

Pollution Loading to Deep Bay WCZ

5.11.10 The water pollution sources in Deep Bay WCZ would mainly include sewage effluent from unsewered developments. **Table 5.17** below gives an estimation on the sewage flow and loading (from unsewered population) generated in the areas to be redeveloped under this Project for the existing and likely future conditions (without this Project). Project areas that will not be disturbed / developed under the Project (e.g. "Green Belt" ("GB") and existing "V" areas) are excluded in this loading estimation. Under the existing condition, sewage flow from existing unsewered developments within the areas to be redeveloped under this Project is estimated to be about 1,616 m^3/day. The methodology and assumptions used for compiling the sewage flow and load are provided in **Appendix 5.3**.

Table 5.17 Existing and Likely Future Sewage Loading to Deep Bay WCZ (Without this Project)

Parameter	Unit	Existing Condition (without this Project), see Notes 1 and 2	Likely Future Condition (without this Project), see Notes 1 and 2
Sewage flow	m^3 /day	1,616	1,895
BOD	kg/day	182	213
SS	kg/day	89	104
NH_3-N	kg/day	46 (see Note 3)	54 (see Note 3)
Org-N	kg/day	see Note 3	see Note 3
TN	kg/day	46	54
TP	kg/day	9	11
E. coli	no./day	2.95×10^{11}	3.46E+11

Note:
(1) Only areas to be redeveloped under this Project are considered in the loading estimation. Project areas that will not be developed under this Project (e.g. "GB" and existing "V" areas) are excluded in the estimation.
(2) All sewage generated from the unsewered areas would be discharged to septic tanks and soakaway system. The loading given in this table represents the residual loading after treatment by septic tanks. Further load reduction due to the soakaway system is excluded in this estimation, which is a conservative approach. With consideration of the soakaway facilities, the actual loading to Deep Bay would be smaller than that shown in this table.
(3) With reference to the approved EIA Report for "Liantang / Heung Yuen Wai Boundary Control Point and Associated Works", the amount of NH_3-N in the effluent of septic tank would be increased (due to the conversion of Org-N to NH_3-N) after the biological reactions within the septic tank but it will not exceed the amount of TN in the effluent, Thus, the Org-N loading is assumed to be negligible.

5.11.11 The future sewage loading (from unsewered areas) to Deep Bay under the "without Project" condition is estimated to be about 1,895 m^3/day (as shown in **Table 5.17** above), which is higher than the existing load solely due to the natural population growth. Assumptions adopted for compiling the likely future loading are given in **Appendix 5.3**. The sewage loading to Deep Bay in the future as shown in **Table 5.17** represents a conservative estimation without considering the effect of any possible new sewerage projects in the area. Since the Government will continue to remove the pollution sources and implement new sewerage projects, the actual pollution loading to Deep Bay in the future (without this Project) could be smaller than that estimated in this EIA.

5.11.12 With implementation of this Project, the area to be developed under this Project will be served with public sewerage, with no unsewered developments.

5.11.13 The existing and likely future non-point source pollution loading from surface run-off generated in the new development area is separately presented in **Tables 5.21 and 5.22** below.

Sewage Overflow or Emergency Bypass due to Capacity Constraints, Aging or Damage of the Sewerage Network

5.11.14 With reference to the Sewerage and Sewage Treatment Implications in **Chapter 6**, sewage flow to be generated from the new developments proposed under this Project will be discharged to the new sewerage system and therefore this Project will not induce any adverse hydraulic impacts upon the existing sewerage system. The new sewerage network will be built with adequate capacity to accommodate the new sewage flow from this Project together with other existing / concurrent projects within the catchment of SW STW. Also, new HSK STW and new SPS with adequate capacities will be provided under this Project to handle the additional sewage flow from this Project and other concurrent projects. Sewage overflow or emergency bypass due to capacity constraints of the sewerage system would not be anticipated.

5.11.15 Ageing or damage of the proposed sewerage system could cause leakage or bursting of the untreated sewage to the nearby receiving waters. Pollutant levels of the receiving watercourses would temporarily increase in case of damage of sewage pipelines or rising mains. In order to prevent the uncontrolled discharge of untreated sewage effluent to surface waters there will be a need to minimise the risk of failure of the sewerage system. Precautionary measures such as using twin rising mains are mentioned in **Section 6.8** and **Section 5.14.1** to minimise the risk of failure of the proposed sewerage system. With proper implementation of the recommended precautionary measures, no adverse water quality impact arising from damage on sewerage system is anticipated.

Sewage Pumping Stations (DP9)

5.11.16 Four new SPS, namely SPS No. 1, SPS No. 2, SPS No.3 and SPS No.4, are proposed under the Project as shown in **Figure 5.5**. The design capacities of the four SPS are summarised in **Table 5.18** below.

Table 5.18 Design Capacities of Sewage Pumping Stations (DP9)

SPS	Estimated Sewage Flow (m^3/day) (1)	Design Capacities (m^3/day)
SPS No. 1	26,641	27,000
SPS No. 2	39,130	39,500
SPS No. 3	10,957	11,000
SPS No. 4	67,943	68,000

Note:
(1) Estimated sewage flow of pumping stations SPS1, SPS2, SPS3 and SPS4 in year 2037/38 as mentioned in **Table 6.14**.

5.11.17 The normal operation of these SPS would actually have beneficial effect through the enhancement of the efficiency of the sewerage system. However, potential water quality impact may arise from emergency overflow / bypass of sewage from the proposed SPS due to pump failure, power supply failure and damage to pressure main or flooding.

5.11.18 Emergency bypass culverts will be built to convey any emergency overflow from the SPS to the nearby watercourses. **Table 5.19** below summarises the tentative discharge point of emergency overflow for each SPS and the downstream receiving water that would potentially affected by the emergency overflow. The tentative alignments of the emergency bypass culverts of the four SPS are shown in **Figure 5.5**.

Table 5.19 Tentative Discharge Points of Emergency Overflow from SPS and Downstream Receiving Water

SPS (see Figure 5.5)	Discharge Point of Emergency Overflow and Downstream Receiving Water (refer to Table 5.3 for detailed description of Receiving Water)
SPS No. 1	Tin Sam Channel, which would then discharge to the TSW Main Channel and finally to the marine water of Deep Bay WCZ.
SPS No. 2	A side branch of Tin Sam Channel, which would firstly drain into Tin Sam Channel and then discharge to the TSW Main Channel and finally to the marine water of Deep Bay WCZ.
SPS No. 3	Tin Sam Channel, which would then discharge to the TSW Main Channel and finally to the marine water of Deep Bay WCZ.
SPS No. 4	Lo Uk Tsuen Tributaries, which would then enter the TSW Main Channel and finally to the marine water of Deep Bay WCZ.

5.11.19 All the selected discharge points of emergency bypass are located at modified watercourses, with man-made concrete embankment and base as mentioned in **Section 9.4.125** of the Ecological Impact Assessment. All these discharge points are also away from sensitive receivers such as fishponds, conservation areas and natural streams etc. Also, no sensitive species of conservation importance (e.g. aquatic dependent species such as fish) was identified in the receiving watercourses.

5.11.20 Under the emergency situation, raw sewage from the SPS proposed under this Project will be discharged to the modified watercourses leading to Deep Bay. SPS with similar emergency discharge arrangement was previously assessed under the approved EIA for SW STW and "Yuen Long and Kam Tin Sewerage and Sewage Disposal Stage 2". The approved EIA for SW STW assessed the water quality impact due to an emergency discharge of raw sewage from Ha Tsuen Sewage Pumping Station of 246,000 m^3/day for a duration of 12 days. Since the emergency discharge assessed in the approved EIA for SW STW would also enter the modified watercourses leading to Deep Bay, similar to the emergency discharge arrangement of this Project, the findings of the approved EIA for SW STW can be applied here.

5.11.21 According to the approved EIA for SW STW, the emergency discharge from Ha Tsuen Sewage Pumping Station would cause an increase in *E. coli*, UIA and TIN in the receiving waters. The baseline conditions for TIN and UIA would be recovered in 5 to 8 days after termination of the emergency discharge. For *E. coli*, its concentration would decrease more rapidly and return to the baseline level in a much shorter period. Since the capacities of the proposed SPS are smaller than that assessed in the approved EIA for SW STW, the time required for *E. coli*, UIA and TIN to return to the baseline value is expected to be shorter. Based on the past experience, breakdown of SPS could be recovered in only hours, which is much shorter than the emergency period of 12 days assessed in the approved EIA for SW STW. Thus, the actual water quality impact that would result from the emergency bypass from the proposed SPS could be much smaller than that predicted in the approved EIA for SW STW.

5.11.22 Since the potential water quality impact arising from overflow, if any, from the proposed SPS would be reversible, no long-term insurmountable water quality impact would be expected from the emergency overflow from the SPS.

5.11.23 In order to minimise the chance of emergency sewage discharge, standby pump would be provided to cater for emergency breakdown or maintenance of the duty pump. Backup power supply in the form of dual / ring circuit power supply or generator would be provided to secure electrical power supply. Hence, the likelihood of complete failure of both "duty" and "standby" power / equipment at two or more SPS together at the same time is considered as an extremely remote situation. The scenario with complete failure of two or more of these SPS together is not expected.

5.11.24 Various precautionary measures to be incorporated in the design of the proposed SPS are recommended in **Section 5.14.2**. With incorporation of the recommended preventive measures, the chance of emergency sewage bypass would be exceptionally rare.

5.11.25 In order to minimise the potential water quality impact in case of emergency discharge, development of a Contingency Plan in the detailed design stage is recommended to deal with the remote occurrence of emergency discharge. An outline of the Contingency Plan is also given in **Section 5.14.3**.

Emergency Discharge from the New HSK STW

5.11.26 As an enhancement measure to minimise the chance of discharging raw sewage to Deep Bay, an emergency bypass culvert will be provided to connect HSK STW to the nearby drainage system in the Deep Bay WCZ. The emergency bypass culvert will allow TSE bypass from the STW to Deep Bay when the NWNT Tunnel / Urmston Road Outfall is out of operation. Under the maintenance / emergency event when the NWNT Tunnel or the Urmston Road Outfall is out of operation, sewage generated from the catchment area will be pumped to HSK STW for treatment and then discharged via the proposed emergency bypass culvert to the nearby drainage channel and then to TSW Main Channel and Deep Bay. With this enhancement measure, the effluent discharged to Deep Bay when the NWNT Tunnel / Urmston Road Outfall are out of operation will be TSE instead of raw sewage. The associated environmental impacts would certainly be lower. The same arrangement has also been adopted for the upgraded SW STW under Agreement 43/2007 (DS).

5.11.27 The tentative alignment of the proposed emergency bypass culvert for HSK STW is indicatively shown in **Figure 5.5**. Phase 1 and Phase 2 of HSK STW would adopt the same emergency bypass culvert. Opportunities for the new HSK STW to use the same emergency bypass culvert of the upgraded SW STW will be explored under the separate Schedule 2 EIA study. Following the same arrangement of the upgraded SW STW, the discharge point of the proposed emergency bypass culvert for HSK STW would be located at an upstream section of Lo Uk Tsuen Tributaries, which would then flow to the TSW Main Channel. The discharge point is selected at modified watercourse, with man-made concrete embankment and base as mentioned in **Section 9.4.125** of the Ecological Impact Assessment. The discharge point is also away from sensitive receivers such as fishponds, conservation areas, natural streams, etc. Also, no sensitive species of conservation importance (e.g. aquatic dependent species such as fish) was identified in the receiving watercourses.

5.11.28 In the event when the SPS are working whilst the treatment at HSK STW is not functioning, untreated sewage will be discharged to the Urmston Road Outfall and then to the marine water of North Western WCZ.

5.11.29 In the emergency event when failure of HSK STW occurs during the maintenance of the NWNT Tunnel / Urmston Road Outfall or when the SPS have broken down, untreated sewage will be discharged to TSW Main Channel and then eventually to the Deep Bay.

5.11.30 The possible emergency discharge scenarios for the new HSK STW are shown in **Table 5.20** below. The same emergency discharge scenarios were also assessed under the approved EIA for SW STW but the design / emergency flow and load of HSK STW would be much lower than that assessed in the approved EIA for SW STW. Hence, the water quality impact induced by the emergency discharges from the new HSK STW is expected to be smaller than that predicted under the approved EIA for SW STW.

Table 5.20 Emergency Discharge Scenarios for the New HSK STW

Scenario	Discharge Route	Receiving Water Body
(A) New HSK STW not functioning and no treatment of the incoming sewage	Raw sewage → NWNT Tunnel → Urmston Road Outfall	North Western waters
(B) NWNT Tunnel / Urmston Road Outfall not in operation	Treated effluent → TSW Main Channel → Deep Bay	Deep Bay waters
(C) New HSK STW and NWNT Tunnel / Urmston Road Outfall not in operation or breakdown of SPS	Raw sewage → TSW Main Channel → Deep Bay	Deep Bay waters

5.11.31 Backup power supply, standby equipment and standby treatment units will be provided in the new HSK STW. Hence, the likelihood of complete failure of both "duty" and "standby" power / equipment at both SW STW and the new HSK STW together at the same time is considered as an extremely remote situation. The scenario with complete failure of both STW together is not expected.

5.11.32 The new HSK STW is a Schedule 2 DP to be assessed under a separate EIA study. The detailed water quality impact of this new HSK STW including the impact due to its emergency discharges will be further investigated in a separate EIA under the EIAO. Due to the sensitive nature of the Deep Bay waters, different design / mitigation measures to minimise the environmental impact of emergency discharge should be thoroughly considered and assessed under the Schedule 2 EIA for HSK STW. For example, provision of emergency storage tank(s) for HSK STW should be considered to minimise the impact under the extreme event of complete failure of both "backup / standby" and "duty" power / plant. Opportunities to temporarily or partially divert the emergency discharge to SW STW for treatment should also be explored to reduce the quantity of emergency discharge.

Surface Run-off from New Developments

5.11.33 Surface run-off to be generated from the Project is known as non-point source pollution. A small amount of oil, grease and grit may be deposited on the surfaces of the road / rail network within the development sites and these pollutants could be washed into the nearby drainage system or watercourses during rainfall events. Surface run-off generated from other paved or developed areas within the development sites may also contain debris, refuse, dust from the roof of buildings and cleaning agents used for washing streets and building façade, which may also affect the quality of the nearby receiving water environment, if uncontrolled.

Areas to be Developed under this Project

5.11.34 Of the total 714 hectares (ha) within the Project area, only about 441 ha will be developed under this Project, with reference to Revised RODP. The remaining Project area of about 273 ha (covering "GB", existing "V" area, etc.) would remain unchanged during the operational phase.

Existing / Future Paved Area (Without this Project)

5.11.35 Within the area to be developed under this Project (of about 441 ha), about 400 ha are already developed / paved areas (including brownfield sites / wasteland / village developments) under the existing situation. The remaining area of about 41 ha consists of greening / unpaved surface (such as plantation, grassland, shrubland, etc.) under the existing condition. Without this Project, the amount of paved / unpaved area in the future is assumed to be the same as that under the existing situation.

Future Paved Area (With this Project)

5.11.36 Based on the Hong Kong Planning Standards and Guidelines (HKPSG) and general guidelines for designing public open space, a minimum 112 ha of greening would be provided under this Project within the area to be developed (of about 441 ha). Thus, the remaining 329 ha would be considered as paved area during the Project operation. The percentages of greening assumed for different land uses of the Revised RODP are given in **Appendix 5.4**.

Non-Point Source Pollution from Surface Run-off

5.11.37 More surface run-off would be generated from the paved area and less from the unpaved area. Assuming 0.9 as the run-off coefficient for paved areas while 0.3 as the run-off coefficient for unpaved surface, the non-point source pollution from surface run-off is quantified in **Table 5.21** below. Project area that will not be developed under the Project (e.g. "GB" / existing "V" areas etc.) is excluded in this estimation. The assumptions used for compiling the non-point source pollution are detailed in **Appendix 5.4**.

Table 5.21 Non-Point Source Pollution from Area to be Developed under the Project

Parameters	Approximate Loading under Existing Condition and Likely Future Condition (without this Project)	Approximate Loading (kg/day) under Likely Future Condition (with this Project)
SS (kg/day)	612	542
BOD (kg/day)	318	282
NH_3-N (kg/day)	3	3
Org-N (kg/day)	17	15
TIN (kg/day), See Note 1	8	8
TN (kg/day)	25	23
TP (kg/day)	3	3

Note 1: Total inorganic nitrogen (TIN) is equal to total nitrogen (TN) minus organic nitrogen (Org-N)

5.11.38 As indicated in the table above, this Project will not cause any net increase in the non-point source pollution discharges into the Deep Bay waters.

5.11.39 Stormwater control measures including Best Management Practices (BMPs), various blue-green infrastructure and Storm Water Pollution Control Plan would be implemented within the development sites to: (1) control erosion and sedimentation; (2) control run-off quantity and quality; (3) eliminate pollutants in point source discharge from drainage outfalls; (4) prevent "first flush" pollution; and (5) eliminate pollutant discharge in stormwater run-off from entering the poor flushing water of Deep Bay, with reference to **Section 2.5** and also described in **Sections 5.14.4 to 5.14.14**. The effects of these stormwater control measures have not been taken into consideration in the quantification of non-point source pollution as given in **Table 5.21** above. It is expected that with proper implementation of the recommended measures, the water quality impact due to the non-point source pollution from the new development area would be minimised.

Surface Run-off from Major Roads (DP1, DP2, DP5, DP6 and DP12)

5.11.40 The identified potential source of impact on water quality due to the operation of road projects would be run-off from the road surfaces. The surface run-off from major roads in the new development area (namely DP1, DP2, DP5, DP6 and DP 12 as defined in **Section 5.1.2**) is assessed in this section. Based on the area of these road works and a run-off coefficient of 0.9 for paved surface as well as the rainfall assumptions given in **Appendix 5.4**, the overall daily run-off generated from major roads (DP1, DP2, DP5, DP6 and DP12) of the Project is estimated to be about 1,494 m^3/day as shown in **Table 5.22** below. It should be noted that the non-point source pollution load estimated in **Table 5.21** have already covered the pollution load generated from surface run-off of these major roads.

Table 5.22 Average Daily Run-off from Major Roads

DP	Area (km^2)	Surface Run-off (m^3/day)
DP1	0.108	369
DP2	0.307	1,049
DP5	0.009	31
DP6 (See Note 1)	0.005	See Note 1
DP9	0.009	31
DP12	0.004	14
Total	**0.442**	**1,494**

Note 1: DP6 is enclosed by decking and therefore the generation of rainwater run-off from DP6 is assumed to be negligible.

5.11.41 The road run-off may contain minimal amount of oil, grease and grit that may cause water quality impacts to the receiving waters in Deep Bay WCZ. Thus, minor non-point source pollution would be expected from these major roads. However, when considering the overall effect due to the Project (which covers all the proposed roads), there will be no net increase in the paved surface (and thus the surface run-off) as compared to the existing condition as discussed in **Sections 5.11.33 to 5.11.38** above.

5.11.42 To minimise the impacts from road run-off, all the road works planned under the Project will be provided with adequate drainage systems and appropriate silt trap, as required. With proper implementation of recommended mitigation measures and best stormwater management practices described in **Sections 5.14.4 to 5.14.14**, adverse impact associated with the discharge of run-off is not anticipated.

Changes of Hydrology and Potential Flooding Risk

Potential Flooding Risk Resulting from Hydrological Changes

5.11.43 The drainage impact arising from the proposed development has been assessed. Based on the review of characteristics of TSW catchment and the arrangement of new drainage network to be built within the Project area, flood retention facilities are proposed to minimise potential flooding risk.

5.11.44 Flood retention facilities will be provided under this Project to protect the low-lying villages, including Shek Po Tsuen, Tin Sam Tsuen, San Lee Uk Tsuen and Kau Lee Uk Tsuen etc., from flooding due to increase in surrounding formation levels as well as long-term climate change. The flood retention facilities would include perimeter interceptor to prevent the ingress of flood water into the low-lying villages and pumping stations to discharge the flood water to the nearby drainage system. Furthermore, flood retention facilities are proposed under the Project at strategic locations as shown in **Figure 5.3**. These flood retention facilities, in form of flood retention lakes or underground retention tanks, will form part of the flood retention facilities to retain stormwater run-off from flowing into the low-lying village areas and surrounding drainage system.

5.11.45 With implementation of the recommended flood protection measures, the potential flooding risk would be minimised. The recommended measures would be completed before the construction works for the development.

Change of Existing Drainage Regime and Water Quality in Tin Shui Wai Main Channel and Inner Deep Bay

5.11.46 TSW Main Channel and its tributaries are a large fresh water system, which spreads across the majority of the Project area as shown in **Figure 5.1**. Thus, stormwater run-off generated in most of the Project area will be eventually discharged into the TSW Main Channel and finally to the Inner Deep Bay. However, as mentioned in **Sections 5.11.33 to 5.11.38** above, this Project will not increase the paved surface and rainwater run-off in the Project area. Thus, no adverse impact on the drainage regime in TSW Main Channel and the overall flow regime in Inner Deep Bay would result from the Project.

5.11.47 All the sewage and wastewater generated from the Project will be properly collected and treated and the treated effluent will be discharged into the NWNT Tunnel for disposal at the North Western WCZ as discussed in **Sections 5.11.1 to 5.11.6** above. Thus, there will be no net increase in pollution loading to the Deep Bay due to the implementation of this Project. Thus, no change in the water quality of Deep Bay would be caused by this Project.

Change of Flow Regime and Hydrology

5.11.48 Several minor watercourses will be removed under the Project. These minor watercourses include: a minor upstream watercourse (namely TSW3) of Lo Uk Tsuen Tributaries; four minor upstream sections (namely TSW1A, TSW1B, TSW1C and TSW1D respectively) that finally join the Tin Sam Channel; and four other minor watercourses (namely TSW2, TSW2A, TSW2B and TSW5 respectively) that finally drain to the TSW Main Channel as summarised in **Table 5.14** above. In addition, a section (namely TSW4) of the Tin Sam Channel will be realigned under the Project. All these small watercourses (namely TSW1A, TSW1B, TSW1C, TSW1D, TSW2 TSW2A, TSW2B, TSW3, TSW4, TSW5) are part of the tributaries of TSW Main Channel. Furthermore, a side branch (namely HHT1) and the most upstream part (namely HH2) of the Hang Hau Tsuen Channel will be removed under the Project. The most upstream part (namely TMR1) of the tributary of Tuen Mun River will also be removed under the Project. Detailed descriptions of these watercourses are given in **Table 5.3** and **Table 5.14** above. The water flow from these watercourses will be diverted to the new drainage system in

the proposed development area. All the watercourses to be removed or diverted are however considered minor water channels only. The main drainage channels within the assessment area including the TSW Main Channel and its key tributaries would remain unchanged within the Project area. The hydraulics of water flow may be changed due to removal or diversion of these minor watercourses, but the impact is expected to be localised and small. No significant change on the flow regime and hydrology within the assessment area is expected.

Change of Ground Water Levels

5.11.49 Dewatering would be performed for the minor watercourses to be removed under this Project. Since most of these minor watercourses are artificial concrete-lined channels and the scale of these watercourses is small, no significant drawdown of groundwater would be expected from the proposed dewatering operation. Dewatering at a few ponds and wet areas would also be required but the scale of all these ponds and wet areas is very small, and hence, no significant groundwater drawdown would be expected from the dewatering work. Overall, no significant change of ground water level in the study area would result from this Project.

Revitalisation and Greening of Drainage Channel Banks

5.11.50 Sections of existing drainage channels including TSW Main Channel, HSK Channel and Tin Sam Channel, running through the Project will be revitalised (e.g. enhancing the greening of the channel banks through the provision of “grasscrete” or similar products). The “grasscrete” or similar products will only be placed on the channel banks above the channel water. No change to the hydrodynamics and water quality would result from the revitalisation and greening works during the Project operation.

Discharge from the Potential District Cooling System (DCS)

5.11.51 Water circulation for the operation of the potential DCS will be in closed circuit. Regular discharge of spent effluent from the proposed DCS is not required under normal operation. During emergency or maintenance condition, spent effluent will be discharged from DCS to the sewerage system. No adverse water quality impact upon the water environment would be expected from the proposed DCS.

Water Reclamation Facility

5.11.52 The water reclamation facility will be installed to treat part of the TSE under tertiary treatment from the new HSK STW for the production of reclaimed water for beneficial uses such as toilet flushing. The reclaimed water to be produced from the water reclamation facility shall meet the specific water quality standards as shown in **Table 5.23**. The water reclamation facility is a Schedule 2 DP and thus the discussion provided here is only preliminary and subject to the future detailed EIA study. The treatment processes to be employed by the water reclamation facility is also subject to confirmation under the separate EIA study.

Table 5.23 Water Quality Standards for Reclaimed Water

Parameter	Unit	Reclaimed Water Quality
E. coli	cfu/100ml	Non detectable
Total Residual Chlorine	mg/L	≥1 (out of treatment system) ≥ 0.2 (at point-of-use)
Dissolved Oxygen	mg/L	≥2
TSS	mg/L	≤5
Colour	Hazen Unit	≤20
Turbidity	NTU	≤5
pH	-	6 to 9
Odour	Threshold odor no.	≤100
BOD_5	mg/L	≤10
Ammonia Nitrogen	mg/L as N	≤1
Synthetic Detergents	mg/L	≤5

5.11.53 Sewage generated from the Project A will be collected by public sewers and subsequently diverted to the STW. The water reclamation facility will not generate extra sewage and therefore will not increase the discharge to the public sewerage system and STW.

5.11.54 In case the operation of the water reclamation facility is suspended under emergency or maintenance situations, the TSE collected from the HSK STW will be discharged back to the sewerage system and eventually to the Urmston Road Outfall in the North Western WCZ. The increase in the pollution load due to the emergency discharge of TSE back to the marine water of the North Western WCZ would be minimal as compared to the emergency discharge cases considered in **Sections 5.11.26 to 5.11.32** above.

5.11.55 The proposed water reclamation facility will not increase the overall sewage load from the Project. In addition, no wastewater / sludge will be discharged into the environment from the proposed water reclamation facility. Hence, no additional source of water quality impact would result from the proposed water reclamation facility.

Proposed HSK Station

5.11.56 All sewage effluent (e.g. from staff and passengers, etc.), wastewater and washed water generated from the proposed HSK Station will be properly collected and discharged to the public sewerage system. No wastewater discharge to the watercourses or nearby water environment would occur from the operation of the proposed HSK Station. The proposed HSK Station is a Schedule 2 DP and thus, the assessment provided here is only preliminary and subject to the future detailed EIA study.

Environmentally Friendly Transport Services and EFTS Depot

5.11.57 No effluent discharge will be generated from the proposed EFTS. Drainage system with silt traps or other suitable pollutant removal devices would be provided to collect the stormwater run-off generated from the EFTS. Under normal condition, stormwater run-off carrying pollutants will not be generated in low rainfall intensity, but increased run-off may occur during heavy rainfall condition. The silt traps and other treatment facilities should be designed with sufficient capacity for the “first flush” flow, which would carry most of the pollutants. The subsequent overland flow generated from rainstorms after the

"first flush" flow is expected to be uncontaminated and thus will be discharged directly to the nearby drainage system bypassing the pollutant removal devices facilities. Prevention of "first flush" pollution in stormwater run-off will be an effective way in controlling pollution at source and to abate pollutants.

5.11.58 Maintenance and cleaning activities as well as canteen (if any) and staff accommodation in the proposed EFTS depot would generate wastewater. Potential water quality impacts would also be generated if accidental spillage occurs from maintenance activities. Surface or washed water run-off generated during the maintenance areas is also potentially contaminated and may pose water quality impact, if uncontrolled.

5.11.59 It would be a required site practice not to directly discharge contaminated surface run-off into the surface channel or nearby water bodies. All the maintenance areas within the depot will be housed or covered to prevent generation of contaminated rainwater run-off. All contaminated surface run-off or wastewater would be collected and diverted to oil interceptors or other appropriate treatment facilities with sufficient design capacities for proper treatment before discharge to the foul sewers.

5.11.60 All waste oils and fuels should be collected and handled in compliance with the Waste Disposal Ordinance. Site drainage should be well maintained and good management practices should be observed to ensure that oils and chemicals are managed, stored and handled properly and do not enter the nearby water streams.

5.11.61 All sewage and wastewater generated from the depot would be properly collected and treated as necessary prior to discharge to the public sewerage system. There will be no direct discharge of wastewater into the stormwater or surface water system.

5.11.62 The EFTS is a potential Schedule 2 DP and therefore the discussion provided above is only preliminary and subject to the future detailed EIA study.

Public Transport Interchange and Petrol Filling Stations

5.11.63 The potential sources of water pollution to be generated from the proposed Public Transport Interchange (PTI) and petrol filling stations would be the potential fuel spillage from the transports and associated contaminated surface run-off or washed water from any floor cleansing activities. The PTI and petrol filling stations would be covered to prevent generation of contaminated rainwater run-off. All contaminated surface run-off or washed water would be collected and diverted to oil interceptor or other appropriate treatment facilities with sufficient design capacities for proper treatment before discharge to the foul sewers.

5.11.64 Fuel spillages should be collected and handled in compliance with the Waste Disposal Ordinance. Site drainage should be well maintained and good management practices should be observed to ensure that oils and chemicals are managed, stored and handled properly and do not enter the nearby water streams.

"Industrial" and "Port Back-up, Storage and Workshop Uses" Sites

5.11.65 Sites 3-43 to 3-52 (except Sites 3-46, 3-48 and 3-49) with a total area of about 13 ha to the west of KSWH along the southwestern boundary of the Project area are designated as "Industrial" ("I") zone as shown in **Figure 2.2**. The "I" zone will provide land for concrete batching plants. The possible types of industrial uses may include warehouses, good storage, printing industry and food manufacturing etc. subject to further detailed study at the detailed design stage.

5.11.66 "Port back-up, Storage and Workshop Uses" ("PBU + SWU") are proposed in the northern Project area at Sites 3-1, 3-4, 3-5, 3-6, 3-7, 3-8, 3-11, 3-13 and 3-14 as shown in **Figure 2.2**. The "PBU + SWU" will provide multi-storey buildings to house some of the affected brownfield operations including the "PBU + SWU" operations. The reserved land in this zone also allows those industries requiring open air operations. The provision of these facilities is subject to further detailed study at the detailed design stage. There will be a potential container back-up and storage area at Sites 3-1, 3-4, 3-5, 3-13 and 3-14, which may be classified as a Schedule 2 DP and thus, the potential water quality impact from the potential container back-up and storage will be subject to detailed assessment to be conducted under a separate EIA study.

Concrete Batching Plants in "Industry" Zone

5.11.67 Concrete batching plants are proposed in the "I" zone. All the works areas including wastewater generating processes and dusty operations of the concrete batching plants will be fully enclosed to avoid loss of dusty materials and generation of contaminated rainwater run-off.

5.11.68 All wastewater generated from the concrete batching plants will be collected, treated, stored and recycled to reduce resource consumption. This includes water used in the concrete batching process, yard washing etc. All spent effluents from the works processes will be collected and diverted to the sedimentation basins with sufficient treatment capacity (to be determined during detailed design stage). The overlying water from the sedimentation basin will be recycled for reuse within the plants. The deposited sediment will be dewatered and the dry matter will require proper disposal off-site. As no water will be discharged outside the boundary of the plants, there will not be any potential water quality impacts associated with the operation of the concrete batching plants.

"Industry" Zone and "Port Back-up, Storage and Workshop Uses" Areas

5.11.69 Industrial wastewater generated in the proposed "I" zone and "PBU + SWU" areas will be properly collected, treated (as required) and then discharged to the foul sewers. Since all the wastewater generating operations are fully enclosed in buildings, no rainwater will be collected. Hence, there will not be any overload of the downstream sewerage system during rainstorms.

5.11.70 It is not possible to estimate the quality of effluent from every type of process that could possibly be carried out in these sites at this EIA stage. However, there is a need for the future operators of these industrial developments to apply to EPD for a discharge licence for discharging effluent from these sites into the foul sewers. The discharge quality is required to meet the water quality standards for effluents discharging to foul sewers and the requirements as specified by the EPD in the discharge licence. The future operators should be required to properly treat the industrial wastewater to meet the WPCO and the TM-DSS before discharging it into public sewerage system. The design capacities and treatment technologies of the on-site WTF, if required, will be determined at the detailed design stage when the detailed design of the industrial processes is available. An outline of an Emergency Response Plan (ERP) for the on-site WTF, if required, is provided in **Section 5.14.20**. By following the ERP, the risk to the surrounding environment from malfunction of the WTF will be minimised.

5.11.71 Domestic wastewater (from toilets) generated in the sites will be discharged directly to the public sewerage system. Clean stormwater will be directed to the nearby stormwater drainage system. Thus, no additional pollution loads will be discharged to the nearby water environment and drainage system due to the operation of these sites.

Surface Discharge from the Open Areas of "I" Zone and "PBU + SWU" Areas

5.11.72 Another water quality concern of the "I" Zone and "PBU + SWU" areas would be the possible contaminated wash-off generated in the open areas of the "I" Zone and "PBU + SWU" areas. This contamination could be the result of accidental spillage.

5.11.73 Perimeter drainage systems will be provided to collect stormwater run-off generated in the open areas of the "I" zone and "PBU + SWU" areas. Under normal operation, rainwater run-off collected in the perimeter drainage system will be diverted to suitable pollutant removal devices (e.g. sedimentation basins and oil interceptors) for treatment. The treated effluent from the pollutant removal devises will be discharged into the nearby stormwater system. Under normal condition, stormwater run-off carrying pollutants will not be generated in low rainfall intensity, but increased run-off may occur during heavy rainfall condition. The pollutant removal devices of the perimeter drainage system should be designed with sufficient capacity for the "first flush" flow, which would carry most of the pollutants. The subsequent overland flow generated from rainstorms after the "first flush" flow is expected to be uncontaminated and thus will be bypassing the pollutant removal facilities and discharged directly to the nearby drainage system. Prevention of "first flush" pollution in stormwater run-off will be an effective way in controlling pollution at source and to abate pollutants under normal situations.

5.11.74 To address the potential water quality concerns under emergency situations, stop-logs will be installed at a suitable location(s) in the perimeter drainage system so that contaminants can be contained in the event of accidental spillage. In the emergency case, stop-logs will be closed to isolate the lot with accidental spillage and prevent it from entering the nearby stormwater system. Contaminated surface water, if any, generated in the lot with accidental spillage will be contained by the stop-logs under the emergency situation. The collected contaminated surface water would then be diverted to the on-site WTF for necessary treatment and then discharged into the foul sewers. To ensure that there is no chance of contaminated run-off leaving the site untreated during high rainfall, the perimeter drainage system will have sufficient capacity (within the channels or at a designated sump) to store any contaminated run-off (spillage plus collected rainwater) from the area isolated by the stop-logs and allow it to be treated at the on-site WTF. If there is any chemical waste collected, the handling and disposal should comply with the requirements under the Waste Disposal Ordinance.

5.11.75 Furthermore, an ERP (to be prepared by the future Operators) will be developed to provide contingency procedures to ensure containment and safe disposal of any accidental spillage or contaminants leaking from the industrial processes. Suitable absorbent materials (e.g. sand or 'vermiculite') shall be kept on site to deal with accidental spillages. An outline of the ERP is provided in **Section 5.14.25**. By following the ERP, the risk to the surrounding environment from any accidental spillage will be minimised.

Refuse Transfer Stations and Refuse Collection Points

5.11.76 The potential sources of water pollution to be generated from the proposed refuse transfer station (RTS) and refuse collection points (RCP) would be the potential spillage of pollutants (e.g. rubbish, dirt, debris, etc.) and associated contaminated surface run-off or washed water from any floor cleansing activities. The RTS and RCP will be housed or covered to prevent generation of contaminated rainwater run-off. All contaminated surface run-off or washed water would be collected and diverted to appropriate treatment facilities for proper treatment before discharge to the foul sewers. Site drainage would be regularly maintained. The proposed RTS is a scheduled 2 DP and thus the discussion above is only preliminary and subject to the future detailed EIA study.

Maintenance of Drainage System

5.11.77 Desilting or maintenance works in watercourses may cause a release of fines and sediment-bound contaminants to the downstream receiving waters. Any required maintenance or desilting work (e.g. to remove any silt, grit or rubbish deposited in the inland water system) should be carried out during periods of low flow in the dry season to isolate the temporary work area from the channel water by barrier to prevent sediment suspension from entering the surrounding water. All debris, grit or rubbish generated from the maintenance activities should be collected and handled in compliance with the Waste Disposal Ordinance. Excavated sediment, if any, generated from the river maintenance activities should be tested and classified in accordance with the ETWB TCW No. 34/2002 for determining the disposal arrangement for the sediment. No direct disposal of the maintenance wastes into the inland watercourses will be allowed.

Fresh Water Service Reservoir (FWSR)

5.11.78 A FWSR is proposed for storage of fresh water for fresh water supply. The water stored in the FWSR will be distributed to the users and there will not be any discharge from the operation of the FWSR. Cleansing of the FWSR, if required, will be conducted by lowering the water in the service reservoir by discharging the fresh or unpolluted water to the stormwater drainage system. Measurement of the residual chlorine level in the FWSR water to be discharged should first be conducted to ensure that any residual chlorine in the FWSR water is properly decayed and lowering of the residual chlorine level in the FWSR water to be discharged is observed prior to any discharge of the FWSR water. The remaining water in the bottom layer (the sludge) will be pumped out and disposed by a licensed waste collector. After the cleansing operation, the cleansing effluent will also be collected by a licensed waste collector and will not be discharged into the stormwater system.

Flushing Water Service Reservoirs (FLWSR) for Use of Reclaimed Water

5.11.79 Two FLWSR are proposed for storage of the reclaimed water from the water reclamation facilities. Reclaimed water stored in the reservoirs will be employed for beneficial uses such as toilet flushing. The FLWSR will be covered to prevent overflow during storm events. Cleansing of the FLWSR, if required, will be conducted by lowering the water in the reservoir by discharging the reclaimed water to the sewerage system. The remaining water in the bottom layer (the sludge) will be pumped out and disposed by a licensed waste collector. After the cleansing operation, the cleansing effluent will also be collected by a licensed waste collector and will not be discharged into the stormwater system. The FLWSR, which form part of the water reclamation system, is a Schedule 2 DP and thus the discussion above is only preliminary and subject to the future detailed EIA study.

5.12 Cumulative Impacts from Concurrent Projects

5.12.1 A list of other potential concurrent projects within the study area is provided in **Table 2.18** of **Chapter 2**. The construction works for all of the concurrent projects as listed in **Table 2.18** would be land-based. Provided that all mitigation measures for construction phase as recommended in the EIA Report would be implemented properly, the potential water quality impact from the construction of the Project is expected to be localised. There would be no significant cumulative construction phase water quality impacts with other potential concurrent projects in the study area.

5.12.2 During the operational phase, the cumulative impacts due to the point source discharges of TSE from the following potential concurrent project has been considered in this water quality impact assessment:

- Yuen Long and Kam Tin sewerage treatment upgrade - Upgrading of San Wai Sewage Treatment Works - Investigation, Design and Construction

5.12.3 The above concurrent project would generate TSE for discharge into the NWNT Tunnel and then eventually to the Urmston Road Outfall for disposal at the marine water of North Western WCZ. Since the TSE generated from this Project would also be discharged to the NWNT Tunnel and the Urmston Road Outfall for disposal at the marine water of North Western WCZ, there would be potential cumulative water quality impacts upon the marine water of North Western WCZ. The potential cumulative water quality impacts from the concurrent discharge has been preliminarily evaluated under this water quality impact assessment with reference to the findings from the past relevant EIA study. Detailed water quality modelling and detailed analysis will be conducted under the separate Schedule 2 EIA for HSK STW to fully assess the cumulative water quality impact due to the TSE discharge in the assessment area.

5.12.4 As discussed in **Section 5.11.38** above, this Project will not cause any increase in the non-point source pollution due to surface run-off arising during the operational phase as compared to the existing or "without Project" scenario. With implementation of the stormwater water control measures including Best Management Practices and blue-green infrastructure, the non-point source pollution would be further minimised and the associated impacts would be localised. Thus, no adverse cumulative water quality impact would be resulted from the non-point source discharge from the Project.

5.13 Mitigation Measures – Construction Phase

Construction Site Run-off and General Construction Activities

5.13.1 Control of potential pollution of nearby water bodies during the construction phase of the Project should be achieved by measures to:

- prevent or minimise the likelihood of pollutants (generated from construction activities) being in contact with rainfall or run-off; and
- abate pollutants in the stormwater surface run-off prior to the discharge of surface run-off to the nearby water bodies.

5.13.2 These principle objectives should be achieved by implementation of the Best Management Practices (BMPs) of mitigation measures in controlling water pollution. The guidelines for handling and disposal of construction site discharges as detailed in the ProPECC PN 1/94 "Construction Site Drainage" should be followed, where applicable. All effluent discharged from the construction site should comply with the standards stipulated in the TM-DSS. The following measures are recommended to protect water quality of the inland and coastal waters, and when properly implemented should be sufficient to adequately control site discharges so as to avoid water quality impacts.

Construction Site Run-off

5.13.3 Surface run-off from construction sites should be discharged into stormwater drains via adequately designed sand / silt removal facilities such as sand traps, silt traps and sedimentation basins. Channels or earth bunds or sand bag barriers should be provided on site to properly direct stormwater to such silt removal facilities. Perimeter channels should be provided on site boundaries where necessary to intercept stormwater run-off from outside the site so that it will not wash across the site. Catchpits and perimeter channels should be constructed in advance of site formation works and earthworks.

5.13.4 Silt removal facilities, channels and manholes should be maintained and the deposited silt and grit should be removed regularly, at the onset of and after each rainstorm to prevent local flooding. Any practical options for the diversion and re-alignment of drainage should comply with both engineering and environmental requirements in order to provide adequate hydraulic capacity of all drains.

5.13.5 With reference to EPD's ProPECC PN 1/94, construction works should be programmed to minimise soil excavation works in rainy seasons (April to September) as far as possible. Hence, the future contractor should plan their works to minimise soil excavation in rainy days. If excavation in soil cannot be avoided in these months or at any time of year when rainstorms are likely, for the purpose of preventing soil erosion, temporary exposed slope surfaces should be covered e.g. by tarpaulin, and temporary access roads should be protected by crushed stone or gravel, as excavation proceeds. Intercepting channels should be provided (e.g. along the crest / edge of excavation) to prevent stormwater run-off from washing across exposed soil surfaces. Arrangements should always be in place in such a way that adequate surface protection measures can be safely carried out well before the arrival of a rainstorm.

5.13.6 Earthworks final surfaces should be well compacted and the subsequent permanent work or surface protection should be carried out immediately after the final surfaces are formed to prevent erosion caused by rainstorms. Appropriate drainage like intercepting channels should be provided where necessary.

5.13.7 Measures should be taken to minimise the ingress of rainwater into trenches. If excavation of trenches in wet seasons is necessary, they should be dug and backfilled in short sections. Rainwater pumped out from trenches or foundation excavations should be discharged into stormwater drains via silt removal facilities.

5.13.8 Open stockpiles of construction materials (e.g. aggregates, sand and fill material) on sites should be covered with tarpaulin or similar fabric during rainstorms.

5.13.9 Manholes (including newly constructed ones) should always be adequately covered and temporarily sealed so as to prevent silt, construction materials or debris from getting into the drainage system, and to prevent stormwater run-off from getting into foul sewers. Discharge of surface run-off into foul sewers must always be prevented in order not to unduly overload the foul sewerage system.

5.13.10 Good site practices should be adopted to remove rubbish and litter from construction sites so as to prevent the rubbish and litter from spreading from the site area. It is recommended to clean the construction sites on a regular basis.

Boring and Drilling Water

5.13.11 Water used in ground boring and drilling for site investigation or rock / soil anchoring should as far as practicable be re-circulated after sedimentation. When there is a need for final disposal, the wastewater should be discharged into stormwater drains via silt removal facilities.

Wheel Washing Water

5.13.12 All vehicles and plant should be cleaned before they leave a construction site to minimise the deposition of earth, mud, debris on roads. A wheel washing bay should be provided at every site exit if practicable and wash-water should have sand and silt settled out or removed before discharging into stormwater drains. The section of construction road between the wheel washing bay and the public road should be paved with backfall to reduce vehicle tracking of soil and to prevent site run-off from entering public road drains.

Acid Cleaning, Etching and Pickling Wastewater

5.13.13 Acidic wastewater generated from acid cleaning, etching, pickling and similar activities should be neutralised to within the pH range of 6 to 10 before discharging into foul sewers.

Effluent Discharge

5.13.14 There is a need to apply to EPD for a discharge licence for discharge of effluent from the construction site under the WPCO. The discharge quality must meet the requirements specified in the discharge licence. All the run-off and wastewater generated from the works areas should be treated so that it satisfies all the standards listed in the TM-DSS. The beneficial uses of the treated effluent for other on-site activities such as dust suppression, wheel washing and general cleaning etc., can minimise water consumption and reduce the effluent discharge volume. If monitoring of the treated effluent quality from the works areas is required during the construction phase of the Project, the monitoring should be carried out in accordance with the relevant WPCO licence.

Construction Works near Watercourses

5.13.15 To minimise the potential water quality impacts from the construction works located near any inland watercourses, the practices outlined in ETWB TC (Works) No. 5/2005 "Protection of natural streams / rivers from adverse impacts arising from construction works" should be adopted where applicable. Relevant mitigation measures are listed below:

- Impermeable sheet piles and cofferdams should be used as required to divert water flow from the construction works area so that all the construction works would be undertaken within a dry zone and physically separated from the watercourses.
- The proposed works should preferably be carried out within the dry season where the flow in the stormwater culvert / water channel / stream is low.
- The use of less or smaller construction plants may be specified in works areas close to the inland water bodies.
- Temporary storage of materials (e.g. equipment, filling materials, chemicals and fuel) and temporary stockpile of construction materials should be located well away from any watercourses during carrying out of the construction works.
- Stockpiling of construction materials and dusty materials should be covered and located away from any watercourses.
- Construction debris and spoil should be covered up and/or disposed of as soon as possible to avoid being washed into the nearby water receivers.
- Construction activities, which generate large amount of wastewater, should be carried out in a distance away from the watercourses, where practicable.
- Mitigation measures to control site run-off from entering the nearby water environment should be implemented to minimise water quality impacts. Surface channels should be provided along the edge of the waterfront within the work sites to intercept the run-off.
- Construction effluent, site run-off and sewage should be properly collected and/or treated.
- Any temporary works site inside the stormwater watercourses should be temporarily isolated, such as by placing of sandbags or silt curtains with lead edge at bottom and properly supported props to prevent adverse impact on the stormwater quality.

- Proper shoring may need to be erected in order to prevent soil / mud from slipping into the inland water bodies.

Revitalisation and Greening of Drainage Channel Banks

5.13.16 The key water quality measure for protection of the revitalised drainage channel water is to avoid polluted site run-off from reaching the revitalised drainage channel water. Relevant mitigation measures should follow the practices outlined in ETWB TC (Works) No. 5/2005 "Protection of natural streams / rivers from adverse impacts arising from construction works" as listed below:

- Impermeable sheet piles and cofferdams should be used as required to divert water flow from the construction works area so that all the construction works would be undertaken within a dry zone and physically separated from the revitalised drainage channel water.
- The proposed works should preferably be carried out within the dry season where the flow in the revitalised drainage channel is low.
- The use of less or smaller construction plants may be specified in works areas close to the revitalised drainage channel.
- Temporary storage of materials (e.g. equipment, filling materials, chemicals and fuel) and temporary stockpile of construction materials should be located well away from the revitalised drainage channel during carrying out of the construction works.
- Stockpiling of construction materials and dusty materials should be covered and located away from the revitalised drainage channel water.
- Construction debris and spoil should be covered up and/or disposed of as soon as possible to avoid being washed into the nearby revitalised drainage channel.
- Construction activities, which generate large amount of wastewater, should be carried out a distance away from the revitalised drainage channel, where practicable.
- Mitigation measures to control site run-off from entering the nearby revitalised drainage channel should be implemented to minimise water quality impacts. Surface channels should be provided along the edge of the revitalised drainage channel within the work sites to intercept the run-off.
- Construction effluent, site run-off and sewage should be properly collected and/or treated.
- Any temporary works site inside the revitalised drainage channel should be temporarily isolated, such as by placing of sandbags or silt curtains with lead edge at bottom and properly supported props to prevent adverse impact on the revitalised drainage channel water.
- Proper shoring may need to be erected in order to prevent soil / mud from slipping into the revitalised drainage channel.

Construction Works in Watercourses / Concrete Flood Storage Pond

5.13.17 The construction method and sequence of the proposed construction in watercourses / concrete flood storage pond for works sites of DP12 should be carefully designed so that all the construction works including any excavation and piling operations would be undertaken within a dry zone and physically separated from the watercourses downstream.

5.13.18 Impermeable sheet pile walls or cofferdam walls or steel casing should be installed to fully enclose the construction works area (including all the excavation and piling works) in the watercourse / pond prior to the commencement of any works in watercourse / pond. Dewatering of the construction works area or diversion of water flow should be undertaken before the construction works to avoid water flow in the construction works area. Silt removal facilities should be used to clarify the effluent generated from the dewatering operation before discharging back to the watercourse / drainage system. Any construction works including excavation and piling activities should be undertaken in a dry zone surrounded by the impermeable sheet pile walls or cofferdam walls or steel casing. Silt curtains should also be deployed around the construction works area inside the watercourse, where practicable, as a second layer of protection to further minimise sediment and contaminant release. All wastewater generated from the piling activities should be regarded as part of the construction site effluent, which should be properly collected and treated as appropriate to meet the standards stipulated in the TM-DSS before disposal. It is recommended that the construction works in watercourses / pond should be undertaken in dry seasons, where practicable, when the water flow is low.

5.13.19 The mitigation measures for construction site drainage as recommended in **Sections 5.13.1 to 5.13.14** and protection measures for construction works near watercourses as detailed in **Section 5.13.15** should also be followed to minimise the potential water quality impacts from the construction in watercourses / pond.

Removal / Diversion of Watercourses

5.13.20 The construction works for removal and diversion of watercourses should be undertaken within a dry zone. Cofferdams or similar impermeable sheet pile walls should be used as necessary to isolate the works areas from the neighbouring waters.

5.13.21 The tentative works sequence for provision of a dry zone for the construction works is described as follows. Construction works at watercourse should be undertaken only after flow diversion or dewatering operation is fully completed to avoid water flow in the works area. Dewatering of watercourse should be performed by diverting the water flow to new or temporary drainage. Where necessary, cofferdams or similar impermeable sheet pile walls should be used to isolate the works areas from neighbouring waters. The permanent or temporary drainage for carrying the diverted flow from existing watercourse to be removed should be constructed and completed before dewatering of that existing watercourse. Construction of all the proposed permanent and temporary drainage should be undertaken in a dry zone prior to receiving any water flow.

5.13.22 The Contractor should provide a dry zone for all the construction works to be undertaken in watercourses and stormwater drainage following the tentative works sequence as described above or using other approved methods as appropriate to suit the works condition. The flow diversion works should be conducted in dry season, where possible, when the flow in the watercourse is low. The wastewater and ingress water from the site should be properly treated to comply with the WPCO and the TM-DSS before discharge.

5.13.23 The site practices outlined in the ProPECC PN 1/94 “Construction Site Drainage” and ETWB TC (Works) No. 5/2005 “Protection of natural streams/rivers from adverse impacts arising from construction works” should be adopted for the proposed demolition or diversion of watercourses where applicable.

Removal / Filling of Ponds and Wet Areas

5.13.24 Construction works at the existing ponds / wet areas should be conducted only after dewatering of these ponds / wet areas is fully completed. The drained water generated from the dewatering of these ponds / wet areas to be removed should be temporarily stored in appropriate storage tanks or containers for reuse on-site as far as possible. Any surplus drained water should be tankered away for disposal at the STW in a controlled manner.

5.13.25 It is recommended to drain only one pond at a time to minimise the potential water quality impact. Dewatering works at ponds / wet areas should be conducted within dry season to minimise the quantity of drained water. No direct discharge of drained water to the stormwater drainage system or marine water should be allowed.

Accidental Spillage

5.13.26 Contractor must register as a chemical waste producer if chemical wastes would be produced from the construction activities. The Waste Disposal Ordinance (Cap 354) and its subsidiary regulations in particular the Waste Disposal (Chemical Waste) (General) Regulation, should be observed and complied with for control of chemical wastes.

5.13.27 Any service workshop and maintenance facilities should be located on hard standings within a bunded area, and sumps and oil interceptors should be provided. Maintenance of vehicles and equipment involving activities with potential for leakage and spillage should only be undertaken within the areas appropriately equipped to control these discharges.

5.13.28 Disposal of chemical wastes should be carried out in compliance with the Waste Disposal Ordinance. The Code of Practice on the Packaging, Labelling and Storage of Chemical Wastes published under the Waste Disposal Ordinance details the requirements to deal with chemical wastes. General requirements are given as follows:

- Suitable containers should be used to hold the chemical wastes to avoid leakage or spillage during storage, handling and transport.
- Chemical waste containers should be suitably labelled, to notify and warn the personnel who are handling the wastes, to avoid accidents.
- Storage area should be selected at a safe location on site and adequate space should be allocated to the storage area.

Sewage Effluent from Construction Workforce

5.13.29 No discharge of sewage to the stormwater system and marine water will be allowed. Adequate and sufficient portable chemical toilets should be provided in the works areas to handle sewage from construction workforce. A licensed waste collector should be employed to clean and maintain the chemical toilets on a regular basis.

5.13.30 Notices should be posted at conspicuous locations to remind the workers not to discharge any sewage or wastewater into the surrounding environment. Regular environmental audit of the construction site should be conducted to provide an effective control of any malpractices and achieve continual improvement of environmental performance on site.

Groundwater from Contaminated Areas, Contaminated Site Run-off and Wastewater from Land Decontamination

5.13.31 Remediation of contaminated land should be properly conducted following the recommendations of Land Contamination Assessment in **Section 8**. Any excavated contaminated material and exposed contaminated surface should be properly housed and covered to avoid generation of contaminated run-off. Open stockpiling of contaminated materials should not be allowed. Any contaminated run-off or wastewater generated from the land decontamination processes should be properly collected and diverted to wastewater treatment facilities (WTF). The WTF shall deploy suitable treatment processes (e.g. oil interceptor / activated carbon) to reduce the pollution level to an acceptable standard and remove any prohibited substances (such as total petroleum hydrocarbon) to an undetectable range. All treated effluent from the wastewater treatment system shall meet the requirements as stated in TM-DSS and should be either discharged into the foul sewers or tankered away for proper disposal.

5.13.32 No direct discharge of groundwater from contaminated areas should be adopted. Prior to any excavation works within the potentially contaminated areas, the baseline groundwater quality in these areas should be reviewed based on the past relevant site investigation data and any additional groundwater quality measurements to be performed with reference to *Guidance Note for Contaminated Land Assessment and Remediation* and the review results should be submitted to EPD for examination. If the review results indicated that the groundwater to be generated from the excavation works would be contaminated, this contaminated groundwater should be either properly treated or properly recharged into the ground in compliance with the requirements of the TM-DSS. If wastewater treatment is to be deployed for treating the contaminated groundwater, the wastewater treatment unit shall deploy suitable treatment processes (e.g. oil interceptor / activated carbon) to reduce the pollution level to an acceptable standard and remove any prohibited substances (such as total petroleum hydrocarbon) to an undetectable range. All treated effluent from the wastewater treatment plant shall meet the requirements as stated in the TM-DSS and should be either discharged into the foul sewers or tankered away for proper disposal.

5.13.33 If deployment of wastewater treatment is not feasible for handling the contaminated groundwater, groundwater recharging wells should be installed as appropriate for recharging the contaminated groundwater back into the ground. The recharging wells should be selected at places where the groundwater quality will not be affected by the recharge operation as indicated in section 2.3 of TM-DSS. The baseline groundwater quality should be determined prior to the selection of the recharge wells, and submit a working plan to EPD for agreement. Pollution levels of groundwater to be recharged shall not be higher than pollutant levels of ambient groundwater at the recharge well. Groundwater monitoring wells should be installed near the recharge points to monitor the effectiveness of the recharge wells and to ensure that no likelihood of increase of groundwater level and transfer of pollutants beyond the site boundary. Prior to recharge, free products should be removed as necessary by installing the petrol interceptor. The Contractor should apply for a discharge licence under the WPCO through the Regional Office of EPD for groundwater recharge operation or discharge of treated groundwater.

Emergency Response Plan for Construction Site Discharges

5.13.34 The following measures should be implemented by the Contractors to minimise the chance of emergency construction site discharge (due to failure of treatment facilities such as sand traps, silt traps, sedimentation basins, oil interceptors etc.):

- Provide spare or standby treatment facilities of suitable capacities for emergency replacement in case damage or defect or malfunctioning of the duty treatment facilities is observed.

- Conduct daily integrity checking of the construction site drainage and treatment facilities to inspect malfunctions, in particular before, during and after a storm event.
- Carry out regular maintenance or desilting works to maintain effectiveness of the construction site drainage and treatment facilities in particular before, during and after a storm event.

5.13.35 An ERP should be developed to minimise the potential impact from construction site discharges under failure of treatment facilities during emergency situations or inclement weather. The ERP should give the emergency contacts to mobilise flood retention facilities and stakeholders to be notified as well as the details of the proposed construction site drainage system and the design and operation of duty and standby treatment facilities. The ERP should also provide the procedures and guidelines for routine integrity checking and maintenance of the drainage system and treatment facilities as well as the emergency response and rectification procedures to restore normal operation of the treatment facilities in case of treatment failure during emergency situation or inclement weather. The Best Management Practices (BMPs) in controlling water pollution arising from the construction activities and an event and action plan with action and limit levels for water quality monitoring should be included in the ERP. The ERP should be submitted to the EPD for approval before commencement of the construction works.

Sewerage Diversion

5.13.36 Construction of the Project would involve diversion of the existing twin 800 mm diameter rising mains along Tin Ying Road. New sewerage facilities for receiving the diverted sewage flow from the existing rising mains should be constructed prior to the commencement of any demolition and construction works at the existing rising mains. All sewage flow running in the existing rising mains along Tin Ying Road should be diverted to the new sewerage system prior to any demolition and construction works at the existing rising mains. No discharge of sewage flow to the environment should be allowed during the sewerage diversion works.

Disposal of Sediment

5.13.37 All excavated materials generated from removal and diversion of watercourses, removal and construction works in ponds and wet areas as well as the proposed bridge pier construction works in watercourses should be collected and handled in compliance with the Waste Disposal Ordinance. Excavated sediment, if any, generated from the excavation activities in watercourses, ponds and wet areas should be tested and classified in accordance with the ETWB TCW No. 34/2002 for determining the disposal arrangement for the sediment. The disposal of excavated sediments should be minimised according to the relevant requirements in the Waste Management Implications in **Section 7.5.27**. No direct disposal of the construction wastes or excavated materials into the stormwater drainage system and marine water should be allowed.

DP1, DP2, DP5, DP6, DP9 and DP12

5.13.38 For DP1, DP2, DP5, DP6, DP9 and DP12, the mitigation measures recommended in **Sections 5.13.1 to 5.13.37** above should be followed, where applicable.

5.14 Mitigation Measures – Operational Phase

TSE Discharge and Sewerage Network

5.14.1 Opportunities should be explored to maximise the use of reclaimed water and reduce the TSE discharge from the HSK STW as far as possible under the further detailed EIA studies for HSK STW. The following precautionary measures are recommended to minimise the risk of failure of the proposed sewerage system:

- Regular inspection, checking and maintenance of the sewerage system;
- Provisions of twin rising mains as backup and to facilitate maintenance and repairing purposes;
- Provisions of leakage collection systems linking to the nearest chamber at its downstream to the rising main for collection of sewage leakage from the damaged rising main;
- Use tankers to store emergency discharge and transport to the STW for disposal in case of both twin rising mains failure; and
- Provisions of spare / standby parts of sewage pipeworks to facilitate maintenance and repairing of equipment.

Sewage Pumping Stations

5.14.2 In order to minimise the chance of emergency sewage discharge, the following precautionary measures are proposed to be incorporated in the design of the SPS:

- A standby pump and screen should be provided to cater for breakdown and maintenance of the duty pump in order to avoid emergency discharge.
- Backup power supply in the form of dual / ring circuit power supply or generator should be provided to secure electricity supply.
- An alarm should be installed to signal emergency high water level in the wet well.
- An emergency storage tank / spare volume of wet well should be provided for the proposed SPS to cater for breakdown and maintenance of duty pump.
- Regular maintenance and checking of plant equipment should be undertaken to prevent equipment failure.
- Twin rising mains system should be provided to facilitate maintenance works and to avoid emergency discharge of sewage.
- A telemetry system to the nearest manned station / plant should be provided so that swift action can be undertaken in case of malfunction of the unmanned facilities.
- A bar screen (with clear spacing of approximately 25 mm) should be provided to cover the lower half of the opening of any emergency sewage bypass which can prevent the discharge of floating solids into receiving waters as far as practicable while ensuring flooding at the facilities would not occur event if the screen is blocked.

5.14.3 A Contingency Plan to deal with the emergency discharges that may occur during operation of the SPS should be developed in the detailed design stage including the following items:

- Locations of water bodies or WSRs in the vicinity of the emergency discharges.

- A list of relevant government departments to be informed and to provide assistance in the event of emergency discharge, including key contact persons and telephone numbers.
- Reporting procedures required in the event of emergency discharges.
- Procedures listing the most effective means in rectifying the breakdown of the SPS in order to minimise the discharge duration.

Best Stormwater Management Practices and Stormwter Pollution Control Plan

5.14.4 With reference to Clause 2.3, Annex 66 of the EIAO-TM, mitigation measures including Best Management Practices (BMPs) to reduce stormwater pollution arising from the Project are as follows.

Design Measures to Control Erosion and Run-off Quantity

5.14.5 Exposed surface shall be avoided within the development sites to minimise soil erosion. The development site shall be either hard paved or covered by landscaping area and plantation where appropriate.

5.14.6 The major water channels and nullahs within the development sites should be retained as far as practicable to maintain the original flow path. The drainage system should be designed to avoid flooding.

5.14.7 Green areas / tree / shrub planting etc. should be introduced within the development site as far as possible including open space and along roadside amenity strips and central dividers, which can help to reduce soil erosion.

Devices and Facilities to Control Sedimentation, Run-off Quality, Prevent "First Flush" Pollution, Eliminate Pollutants in Point Source Discharge from Drainage Outfalls and Eliminate Pollutant Discharges into the Poor Flushing Water of Deep Bay WCZ

5.14.8 Screening facilities such as standard gully grating and trash grille, with spacing which is capable of screening large substances such as fallen leaves and rubbish should be provided at the inlet of drainage system.

5.14.9 Road gullies with standard design and silt traps and oil interceptors should be incorporated during the detailed design to remove particles present in stormwater run-off, where appropriate.

5.14.10 Evergreen tree species, which in general generate relatively smaller amount of fallen leaves, should be selected where possible.

Administrative Measures to Control Sedimentation, Run-off Quality, Prevent "First Flush" Pollution, Eliminate Pollutants in Point Source Discharge from Drainage Outfalls and Eliminate Pollutant Discharges into the Poor Flushing Water of Deep Bay WCZ

5.14.11 Good management measures such as regular cleaning and sweeping of road surface / open areas are suggested. The road surface / open area cleaning should also be carried out prior to occurrence rainstorm.

5.14.12 Manholes, as well as stormwater gullies, ditches provided at the development sites should be regularly inspected and cleaned (e.g. monthly). Additional inspection and cleansing should be carried out before forecast heavy rainfall.

Blue-green Infrastructure to Control Sedimentation and Run-off Quantity

5.14.13 With reference to **Section 2.5**, various blue-green infrastructure have been proposed under this Project to reduce the drainage loading to the drainage system. They include:

- Provision of bioswales, where practicable at roadside, to convey stormwater and provide removal of coarse and medium sediments. As the water is transported along the bioswales, it is treated to remove pollutants and the cleaned water can then be discharged into the receiving water bodies or retained for non-potable reuse, e.g. irrigation.
- Rainwater harvesting should be implemented within the development site, where possible, to collect rainwater from building roofs, podiums, walkway canopies and other built structures for reuse as an alternative water source e.g. irrigation. The system should meet the prevailing WSD guidelines.
- Porous paving material should be used, where practical to increase stormwater infiltration and improve groundwater recharge and reducing flooding from surface run-off.

Stormwater Pollution Control Plan

5.14.14 A Stormwater Pollution Control Plan should be developed for potential polluting facilities to prevent or minimise the potential of pollutants coming into contact with rainwater or run-off. The plan shall incorporate details such as locations, sizes and types of measures / installations and the BMPs.

"Industrial" and "Port Back-up, Storage and Workshop Uses" Sites

Concrete Batching Plants in "Industry" Zone

5.14.15 All the works areas including wastewater generating processes and dusty operations of the concrete batching plants should be fully enclosed to avoid generation of contaminated rainwater run-off.

5.14.16 All wastewater generated from the concrete batching plants should be collected, treated, stored and recycled to reduce resource consumption. All spent effluents from the works processes should be collected and diverted to sedimentation basins with sufficient treatment capacity. The overlying water from the sedimentation basins should be recycled for reuse within the plants. The deposited sediment should be dewatered and the dry matter should be properly disposed off-site.

5.14.17 Stormwater or rainwater run-off is uncontaminated and shall be physically separated from the wastewater streams and spent effluent generated from the works processes of the concrete batching plants.

Multi-storey Industrial Buildings in "Industry" Zone and "Port Back-up, Storage and Workshop" Areas

5.14.18 Industrial wastewater generated in the proposed multi-storey buildings in the "I" zone and "PBU + SWU" areas should be properly collected, treated (as required) and then discharged to the foul sewers.

5.14.19 The future operators of the industrial developments should apply to EPD for a discharge licence for discharging effluents from the multi-storey industrial buildings into the foul sewers. The discharge quality should meet the water quality standards for effluents discharging to the foul sewers and the requirements as specified by the EPD in the discharge licence. The future operators should be required to properly treat the industrial wastewater to meet the WPCO and the TM-DSS before discharging it into public sewerage system. The design capacity and treatment technologies of the on-site WTF

should be determined during the detailed design stage when detailed design of the industrial processes is available.

5.14.20 An ERP should be developed by the future Plant Operators to deal with emergency situations caused by malfunctioning of the on-site WTF. The ERP for the on-site WTF should cover the following:

- Contact personnel and the means to contact.
- Procedures to initiate emergency repairs.
- Procedures to temporarily divert the incoming effluent to any designated temporary holding facility.
- Procedures to partially / fully treat effluents at an alternative treatment facility.

5.14.21 The EPR should be developed in the detailed design stage when the detailed design of the WTF and industrial processes is available. Domestic wastewater (from toilets) generated in the multi-storey buildings should be discharged to the public sewerage system.

Surface Discharge from the Open Areas of "Industry" Zone and "Port Back-up, Storage and Workshop" Areas

5.14.22 Perimeter drainage systems should be provided to collect stormwater run-off generated in the open areas of the "I" zone and "PBU + SWU" areas. Under normal operation, rainwater run-off collected in the perimeter drainage system should be diverted to suitable pollutant removal devices (e.g. sedimentation basins and oil interceptors) for necessary treatment and then discharged into the nearby stormwater system. The pollutant removal devices of the perimeter drainage system should be designed with sufficient capacity for the "first flush" flow, which would carry most of the pollutants. The subsequent overland flow generated from rainstorms after the "first flush" flow should be bypassing the pollutant removal facilities and discharged directly to the nearby drainage system.

5.14.23 Stop-logs should be installed at a suitable location(s) in the perimeter drainage system so that contaminants can be contained in the event of accidental spillage. Under emergency situations, the relevant stop-logs should be closed to isolate the lot with accidental spillage and prevent it from entering the nearby stormwater system. Contaminated surface water, if any, generated in the lot with accidental spillage should be contained within the site by the stop-log system. The collected surface water should be diverted to the on-site WTF for necessary treatment and the treated effluent from the on-site WTF should be discharged into the foul sewers. The effluent of the on-site WTF should meet the water quality standards and the requirements of the discharge licence for effluents discharging into the foul sewers. To ensure that there is no chance of contaminated run-off leaving the site untreated during high rainfall, the perimeter drainage system should have sufficient capacity (within the channels or at a designated sump) to store any contaminated run-off (spillage plus collected rainwater) from the area isolated by the stop-logs and allow it to be treated at the on-site WTF. If there is any chemical waste collected, the handling and disposal should comply with the requirements under the Waste Disposal Ordinance.

5.14.24 Good management and housekeeping measures such as regular cleaning and sweeping of surface / open areas are recommended. All stormwater drainage and pollutant removal devices should be regularly inspected and cleaned (e.g. weekly). Additional inspection and cleaning should be carried out before forecast heavy rainfall.

5.14.25 An ERP should be developed by the future operators at the detailed design stage to deal with emergency situations of accidental spillage. The ERP should cover the following:

- Contact personnel and the means to contact.
- Procedures to contain contaminants and prevent their escape and/or dispersion, e.g., through closing the stop-logs to isolate in the lot's perimeter drainage system from the surrounding stormwater drainage system.
- Procedures to divert / transport the contaminated materials to a designated temporary storage area or appropriate treatment facility.
- Procedures to clear up the lot and/or perimeter drainage system prior to opening the stop-logs.

Control of Operational Site Effluents

5.14.26 The practices outlined in ProPECC PN 5/93 "Drainage Plan subject to Comments by Environmental Protection Department" should be adopted where applicable for handling, treatment and disposal of operational stage effluent. Drainage outlets provided in covered areas, such as covered railway station, covered electricity substation, covered podiums and other roofed areas, should be discharged to foul sewers.

5.14.27 Drainage serving covered PTI, covered petrol filling stations, covered RTS / RCPs and covered EFTS depot should be connected to foul sewers. Sedimentation facilities, petrol interceptors or other appropriate wastewater treatment system should be provided to treat the wastewater or surface run-off generated in these facilities as necessary to meet the discharge standards as stipulated in the TM-DSS prior to the discharge of these effluents to the public sewers.

Maintenance of Drainage System

5.14.28 For maintenance of stormwater drainage system, reference should be made to ETWB TC (Works) No. 14/2004 "Maintenance of Stormwater Drainage Systems and Natural Watercourses" where applicable. The circular sets out the departmental responsibilities for the maintenance of stormwater drainage systems and natural watercourses in government and private lands. Any required maintenance or desilting work (e.g. to remove any silt, grit or rubbish deposited in the inland water system) should be carried out during periods of low flow in the dry season to minimise impacts on downstream water quality and sediment suspension.

DP1, DP2, DP5, DP6, DP9 and DP12

5.14.29 Mitigation measures recommended in Sections 5.14.2 and 5.14.3 above are applicable to DP9.

5.14.30 Mitigation measures recommended in Sections 5.14.4 to 5.14.12 above are applicable to DP1, DP2, DP5, DP6 and DP12.

5.15 Evaluation of Residual Environmental Impacts

5.15.1 With proper implementation of the recommended mitigation measures, no unacceptable residual water quality impact would be anticipated from the Project during construction and operational phases.

5.16 EM&A Requirements

5.16.1 Water quality monitoring at all major watercourses identified within the Project area is recommended to be carried out before, during and after the construction phase. Details of the recommended water quality monitoring requirements are provided in the stand-alone EM&A Manual for the Project. It is recommended that regular site inspections during the construction phase should be undertaken to inspect the construction activities and works areas in order to ensure the recommended mitigation measures are properly implemented.

5.17 Environmental Acceptability of Schedule 2 Designated Projects

5.17.1 An application for EPs would be submitted under this EIA for DP1, DP2, DP5, DP6, DP9 and DP12 (refer to **Figure 1.3** and **Section 5.1.2**).

Sewage Pumping Station (DP9)

5.17.2 With proper implementation of the recommended mitigation measures and Best Management Practices (BMPs) for construction activities (as detailed in **Sections 5.13.1 to 5.13.37** above), as well as the precautionary design measures and Emergency Response Plan (ERP) to minimise / avoid emergency bypass (as described in **Sections 5.14.2 and 5.14.3** above), no unacceptable water quality impact would be resulted from the proposed SPS (DP9).

New Roads (DP1, DP2, DP5, DP6 and DP12)

5.17.3 With proper implementation of the recommended mitigation measures and BMPs for construction activities (as detailed in **Sections 5.13.1 to 5.13.37** above), as well as the mitigation measures and BMPs to reduce pollution arising from the surface water run-off during the operational phase (as described in **Sections 5.14.4 to 5.14.12** above), no unacceptable water quality impact would be resulted from the proposed roads (DP1, DP2, DP5, DP6 and DP12).

Other DPs

5.17.4 There will be separate EIA studies to assess the following Schedule 2 DPs (refer to **Figure 1.2**):

- DP3 - Construction of new West Rail HSK Station (Site 4-34)
- DP4 - Construction of EFTS – subject to further review
- DP7 - Construction of a new container back-up and storage area (Site 3-1, 3-4, 3-5, 3-13 and 3-14) - subject to further review
- DP8 - Construction of new HSK STW (Site 3-26 and part of existing SW STW)
- DP10 - Construction of Flushing Water Service Reservoirs for reuse of reclaimed water at Tan Kwai Tsuen and Fung Kong Tsuen (Sites 3-3 and 5-40)
- DP11 - Construction of one RTS (Site 3-12)

5.17.5 The detailed water quality impact of these Schedule 2 DPs including the water quality impacts from the new HSK STW will be further investigated in separate EIA studies under the EIAO. The EM&A requirements for these Schedule 2 DPs will also be provided under separate EIA studies.

5.17.6 In particular, detailed water quality modelling and detailed analysis will be conducted under the separate Schedule 2 EIA to assess the water quality impact due to the discharges from the proposed new HSK STW under both normal operation and emergency situations.

5.18 Conclusion

Construction Phase

5.18.1 Water quality impacts from the construction works are associated with the general construction activities, construction site run-off, accidental spillage, and sewage effluent from construction workforce. The site practices as outlined in the ProPECCPN 1/94 "Construction Site Drainage" and the ETWB TC (Works) No. 5/2005 "Protection of natural streams / rivers from adverse impacts arising from construction works" are recommended to minimise the potential water quality impacts from the construction activities. Proper site management and good site practices are also recommended to ensure that construction wastes and other construction-related materials would not enter the nearby watercourses. Sewage effluent arising from the construction workforce would be handled through provision of portable toilets. Water quality monitoring and regular site inspection will be implemented for the construction works to ensure that the recommended mitigation measures are properly implemented.

5.18.2 An ERP is recommended to minimise the potential water quality impact from construction site discharges under failure of treatment facilities during emergency situations or inclement weather.

5.18.3 With the implementation of the recommended mitigation measures, the construction works for the Project would not result in unacceptable impacts on water quality.

Operation Phase

5.18.4 All sewage generated from the Project will be discharged to the public sewerage system and diverted to HSK STW for proper treatment. All the TSE from HSK STW will be pumped to the North West New Territories Effluent Tunnel for discharging to the Urmston Road Outfall for proper disposal at the North Western WCZ. Hence, no net increase in the pollution loading to the Deep Bay waters would be induced by the Project sewage effluent. The Project would actually have water quality beneficial effect by providing new sewerage to the existing unsewered areas and thus reducing the pollution loading to Deep Bay.

5.18.5 In view of the potential adverse effect of emergency sewage bypass and sewage leakage on the quality of the nearby watercourses, various precautionary measures are proposed to be incorporated in the design of the SPS and rising mains to avoid emergency bypass and leakage of sewage to the maximum practicable extent. A Contingency Plan is also recommended to deal with the remote occurrence of emergency discharge. With the incorporation of the precautionary measures and Contingency Plan as recommended in this EIA, the possibility of emergency sewage bypass and sewage leakage would be remote and the potential water quality impacts in the unlikely event that an overflow / leakage does occur would be minimised.

5.18.6 Another source of potential impact during the operational phase will be the run-off or non-point source pollution from road surfaces and developed areas. Stormwater control measures including adequate stormwater drainage system with suitable pollutant removal devices, blue-green infrastructure and best stormwater management practices are recommended for the Project to minimise the non-point source pollution. With proper implementation of the recommended mitigation measures, it is anticipated that the water quality impacts associated with the non-point source discharge from road surfaces and developed areas would be minimised.

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